# Chapter 6

# MVA-Based Offline Preselection

As discussed in chapter 3, the HLT is one of the first steps in selecting data and MC events for analysis. Once a decision has been rendered by the HLT, the event is read out and saved in either the appropriate data stream or in the corresponding MC sample. Once these files are read in by the analysis software – in this case HiggsDNA as described in section 3.4.4 – further selection cuts are made to create a final analysis sample on which measurements can be made.

## 6.1 Current Diphoton Preselection

The current diphoton selection is created by direct analogy to the standard diphoton HLT; firstly, $p_T$, showershape, and isolation cuts are made on individual photons in an event, then the single photons are paired together and cuts are made on the resulting diphoton. In the case of the HLT, a simple diphoton mass cut is made. At the offline analysis level, however, the diphoton cuts are made on $p_T/M_{\gamma\gamma}$.

The $H \to \gamma\gamma$ preselection is described in detail in [12]. First, loose cuts of $p_T > 20GeV/c$, $R_9 > 0.5$, $H/E < 0.08$, and $idMVA > 0.9$ are applied to each photon candidate. Additionally, the $|\eta_{sc}|$ is constrained to be below 2.5, and the gap region between the EB and EE is excluded (1.444 $< |\eta_{sc}| <$ 1.556). Additionally, the electron veto is required to exclude photons origininating from electron tracks. Beyond that, the remaining isolation and showershape cuts are only applied

**Table 6.1.** Table showing the cuts applied by the standard preselection for $H \to \gamma\gamma$.

| **Region** | $R_9$ | **H/E** | **idMVA** | $\sigma_{i\eta i\eta}$ | **Pho Iso** (GeV) | **Track Iso** (GeV) |
|---|---|---|---|---|---|---|
| Barrel | [0.5,0.85] | $< 0.08$ | $' > 0.9$ | $> 0.015$ | $< 4.0$ | $< 4.0$ |
| | $> 0.85$ | $< 0.08$ | $> 0.9$ | – | – | – |
| Endcap | [0.5,0.9] | $< 0.08$ | $> 0.9$ | $> 0.035$ | $< 4.0$ | $< 4.0$ |
| | $> 0.9$ | $< 0.08$ | $> 0.9$ | – | – | – |

on low-$\eta$ photons. Above $R_9 = 0.85$ in the barrel and $R_9 = 0.90$ in the endcap, no further preselection cuts are applied. For photons below these $R_9$ thresholds, cuts are applied on the $\sigma_{i\eta i\eta}$, photon isolation, and track isolation. Finally, the electron veto (described in REF) is required for each photon candidate. This veto decision is designed to identify and remove energy signatures associated with a track, which are likely to be electrons.

Once these single photon preselection cuts are applied, all remaining photon canidates are paired together into a collection of diphotons. For each pair, the higher of the two photons must satistfy $p_T > 30GeV/c$. Paired with the baseline requirement of $p_T > 20GeV/c$, these cuts closely mimic the 30 and 22 GeV/c cuts applied at the trigger level. Then, the $p_T/M_{\gamma\gamma}$ cuts are applied to the diphoton pairs. At this point, many of the possible photon combinations have been removed. A final diphoton combination must be selected, however, as HiggsDNA is designed to output a single photon pair of photons. The remaining pairs in the event are sorted by their diphoton $p_T$, and the pair with the highest value is written out into the event [12]. These cuts, in their totality, are shown in table 6.1

There is no explicit $M_{\gamma\gamma}$ cut applied within the preselection, as the possibility for high- and low-mass Higgs analysis is left open. For the purposes of comparing this preselection to the new MVA-based selection, only the standard mass Higgs is considered. So, a loose cut of $90 < M_{\gamma\gamma} < 180GeV/c^2$ is applied to all resulting diphotons.

## 6.2 Development of MVA-Based Offline Preselection

Since the offline selection is designed to cover the HLT used, it is clear that applying the preexisting preselection described above on top of the MVA-HLT is a suboptimal strategy. In fact, since these cuts mirror the standard HLT quite closely, it is expected that applying the standard preselection on top of the MVA-HLT would likely reduce or even entirely remove the signal efficiency gains obtained by the new trigger. As a result, an alternative set of preselection cuts must be proposed that allows the trigger-level gains to be realized at the offline analysis level.

Just as the standard preselection cuts are designed to not only cover the standard HLT but work in much the same way, a new preselection can be created with maximum similarity to the MVA-HLT. One major complication of this is that the same BDTs cannot be used for both the trigger and offline selection. Since the reconstruction differs for the NanoAOD vs the RAW, the input variable distributions will be different enough that the models are not directly comperable. Instead, a new BDT-based preselection framework must be developed and its coverage of the MVA-HLT must be analyzed.

### 6.2.1 Creation of Single Photon BDTs

#### A $\gamma + Jet$ BDT

As discussed in section 6.1, there already exists an MVA within the standard preselection. While this idMVA is now a centralized part of the NanoAOD format, historical preselections used an MVA trained of $\gamma + Jet$ MC (REF?). Events in $\gamma + Jet$ MCs will contain one real (prompt) photon and one jet that looks like a photon, termed a ”fake”. This MC provides a very easy-to-use training sample, as one can simply use the prompt $\gamma$ as the signal photon and the fake as the background photon. This standard $\gamma + Jet$ idMVA is usually only used to make relatively loose cuts as part of the much broader preselection cuts as described above.

The original idMVA was trained using many of the variables from the preselection as

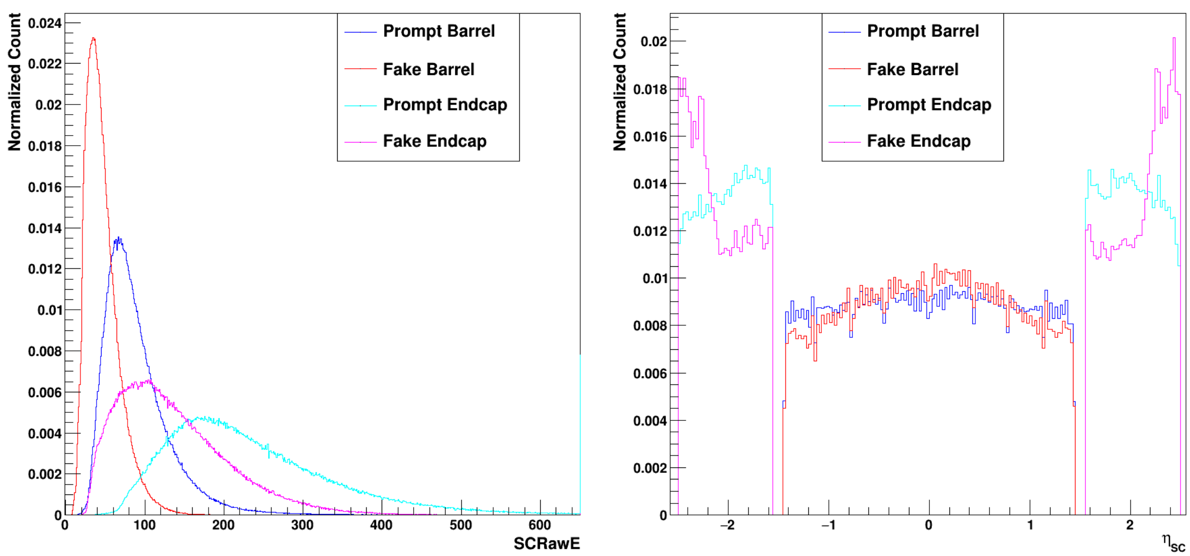
**Table 6.2.** Table of input variable names for the offline $\gamma+Jet$ MVA, and a short description of each. The final two variables are only used for training the endcap.

| Variable Name | Description |
|---|---|
| Raw Energy | The total energy deposited in the 5x5 supercluster |
| $R_9$ | The proportion of the energy contained in the central 3x3 crystals relative to the raw energy |
| $S_4$ | The proportion of the energy contained in the central 2x2 crystals relative to the raw energy |
| $\sigma_{i\eta i\eta}$ | The covariance in terms of crystal spacing ($i_\eta$) |
| $cov(i\eta i\phi)$ | The covariance between the $\eta$ and $\phi$ coordinates |
| $\eta_{SC}$ | The eta coordinate of the center of the supercluster |
| $\eta$ Width | The supercluster width in $\eta$ |
| $\phi$ Width | The supercluster width in $\phi$ |
| H/E | The ratio of the total supercluster energy measured in the HCal to that measured in the ECal |
| PhoIso03 | The photon isolation |
| Charge Iso w.r.t. Worst Vertex | The charged isolation for the worst vertex determined by the vertexing algorithm |
| Charge Iso w.r.t. Chosen Vertex | The charged isolation for the chosen vertex determined by the vertexing algorithm |
| $\rho$ | The event energy density |
| esEffSigmaRR | temp |
| esEnergyOverRawE | temp |

input variables. These same input variables were used as a starting point for a new $\gamma+Jet$ MVA, with the addition of H/E. By adding this variable to the inputs, the new $\gamma+Jet$ MVA contains the same information available to the standard preselection described above. Additionally, this allows the offline selection model to more closely mirror the MVA-HLT. The input variables used are shown below in table 6.2. As with the MVA-HLT, the barrel and endcap models are trained separately, with individual photons selected based on their location in the detector.

As with the trigger level BDT, the sparation between the signal and background in the various input variables can be examined. For the subsequent plots (6.1-6.7), the prompt and fake will be shown for both the barrel and endcap, as the trainings are separated. The prompt photons will be shown in dark and light blue for the barrel and endcap, respectively. The fake distributions will be shown in red and pink for the barrel and endcap, respectively. All the input variable distributions have been normalized to 1, so that their shapes can be directly compared.

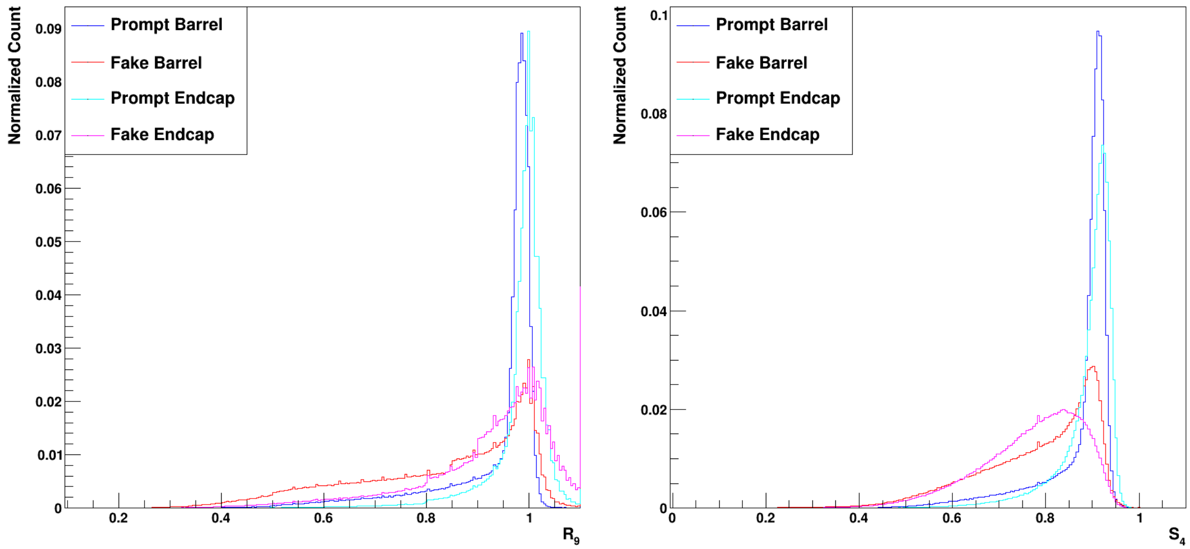
The SC raw energy, in 6.1a, shows rather clear separation for both the barrel and endcap. It can be seen that the prompt photons tend, on average, to have higher total supercluster energy than the fake photons. The separation in $\eta_{SC}$, however, is primarily seen in the endcap, where the fake photons tend towards the higher values. Since the BDT works by creating correlations between variables, and the $\eta_{SC}$ is a strongly kinematic-dependent variable, it will still be shown to be a relatively powerful discrimination variable.




**(a)** Raw Energy **(b)** $\eta_{SC}$

**Figure 6.1.** Plots of the Supercluster raw energy and $\eta_{SC}$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

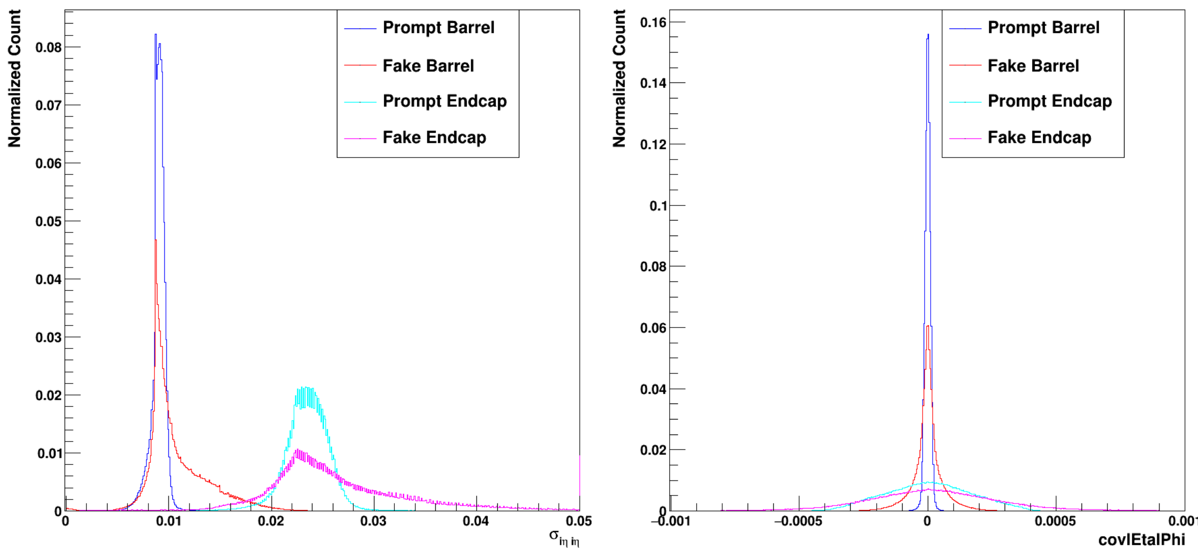
Next, the energy ratios, $R_9$ and $S_4$ can be examined. It is expected that a real (prompt) photon will have relatively well-localized energy; since the photon is a relatively clean signal in the detector, it is expected that most of the energy will be deposited in the center of the supercluster. It is clear that this is the case for both the barrel and endcap, as seen in figure 6.2. Since the fake photon corresponds to a jet, it is expected that the energy deposition will be more spread in the supercluster, which can also be seen in the plots. As a result, $R_9$ and $S_4$ can be relativelty powerful discrimination variables.




**(a)** $R_9$ **(b)** $S_4$

**Figure 6.2.** Plots of the $R_9$ and $S_4$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

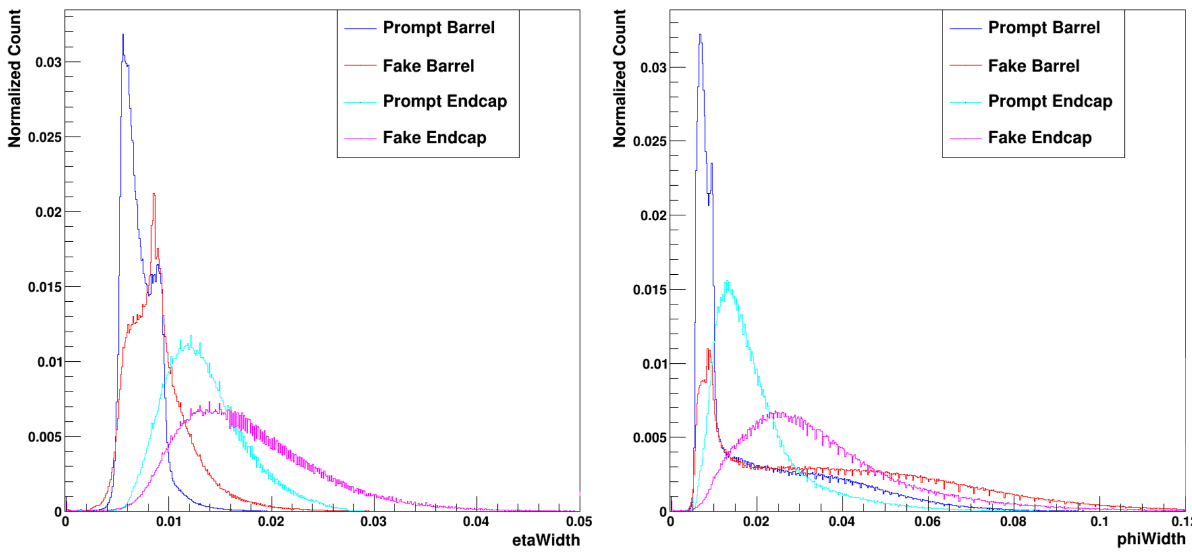
The next showershape variables, $\sigma_{i\eta i\eta}$ and cov$(i\eta i\phi)$, provide information about the energy depositions in terms of the crystal spacing. $\sigma_{i\eta i\eta}$ quantifies the standard deviation of the energy depisited in terms of the $\eta$ crystals in the supercluster. Cov$(i\eta i\phi)$, on the other hand, quantifies the covariance between the energy deposited in the $\eta$ and $\phi$ directions. Since these measurements both depend on the crystal size and spacing, their values differ considerably between the varrel and endcap. This can be seen clearly in figure 6.3. In the barrel, the $\sigma_{i\eta i\eta}$ and cov$(i\eta i\phi)$ both prove to be very good discrimination variables; the fake photons tend to have considerably larger spread in both measures compared to the prompt photons. In the endcap, the signal $\sigma_{i\eta i\eta}$ is still clearly well-separated from the background, though the separation in cov$(i\eta i\phi)$ is much smaller.




**(a)** $\sigma_{i\eta i\eta}$ **(b)** $\mathrm{cov}(i\eta i\phi)$

**Figure 6.3.** Plots of the $\sigma_{i\eta i\eta}$ and $\mathrm{cov}(i\eta i\phi)$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

Similarly to the $\sigma_{i\eta i\eta}$ and $\mathrm{cov}(i\eta i\phi)$, the $\eta$ and $\phi$ widths are powerful showershape variables that provide some measure of how well-localized the energy deposition is. It is expected that these values are different for the barrel and endcap, as the crystal size and spacing are different in the two different regions of the detector. This can be seen in both plots in figure 6.4. Additionally, the separation between prompt and fake photons can be clearly seen, especially in $\eta$ width.

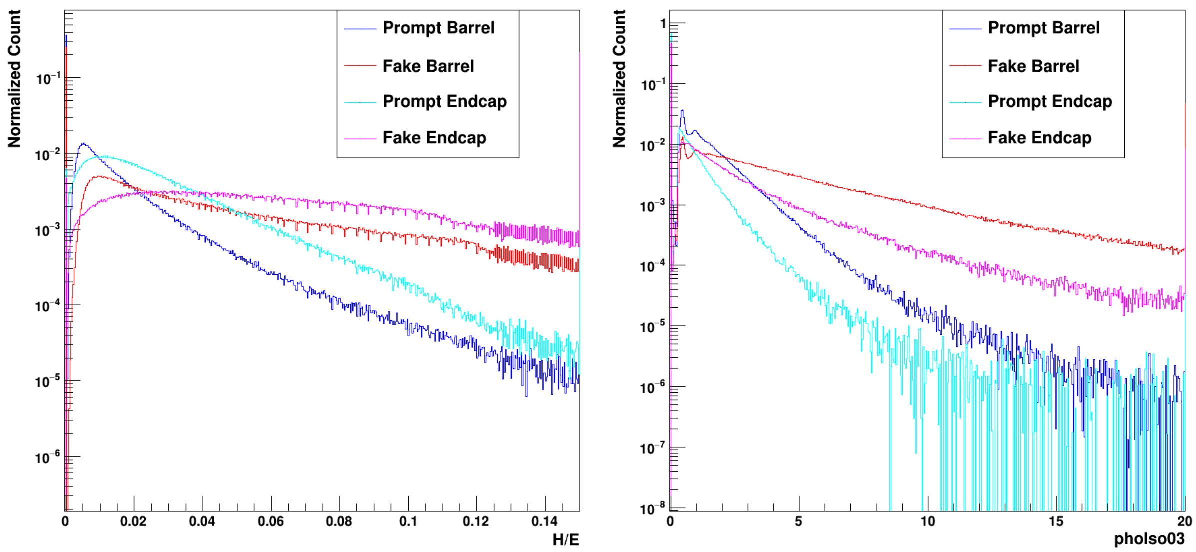



**(a)** $\eta$ Width

**(b)** $\phi$ Width

**Figure 6.4.** Plots of the $\eta$ and $\phi$ widths. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The measure of the HCal energy vs the ECal energy, H/E, provides a powerful measure of whether the energy deposition more closely matches an electromagnetic process – such as a photon or electron – or a hadronic process such as a jet. It is expected that most photons will have a value consistent with 0, as they should deposit all of their energy into the ECal. There is, of course, some level of energy leakage that ensures that even photons often do not have an H/E of identically 0. Clearly, however, the prompt photons in both the barrel and endcap strongly peak near 0 and fall off rather quickly below this. Note that figure a is a log plot, so that the secondary peak below 0 is down by more than a factor of 10 relative to the primary peak at 0. Since the fake photons are jet signatures designed to mimic a prompt signature, they still exhibit a strong peak near 0. It can be seen, however, that the distribution extends to much higher values of H/E overall.

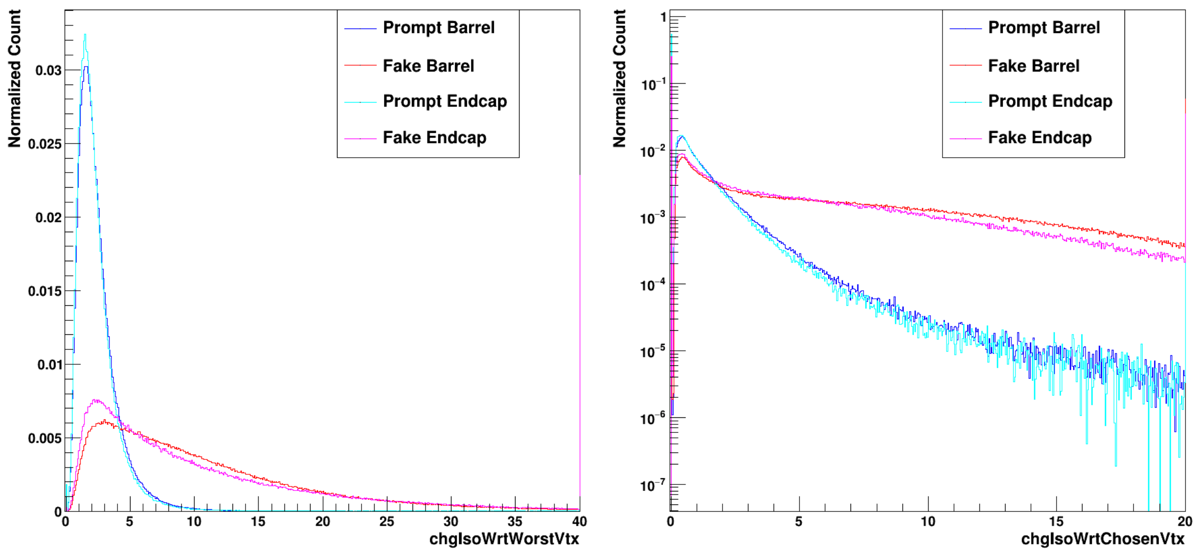



**(a)** H/E

**(b)** Photon Isolation

**Figure 6.5.** Plots of the H/E and photon isolation. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The charge isolation variables are the next input variables, and represent the isolation energy (in GeV) relative to a specific vertex. The vertexing described in section 3.4.1 provides multiple vertices at the NanoAOD level. Isolations are measured relative to both the worst possible vertex (figure 6.6a), and the final chosen vertex (figure 6.6b). While the chosen vertex often has 0 isolation energy, the worst vertex very rarely does. In both cases, however, the separation between signal and background is clear for both the barrel and endcap. The charge isolation w.r.t. worst vertex, specifically, shows very clear and powerful spearation.

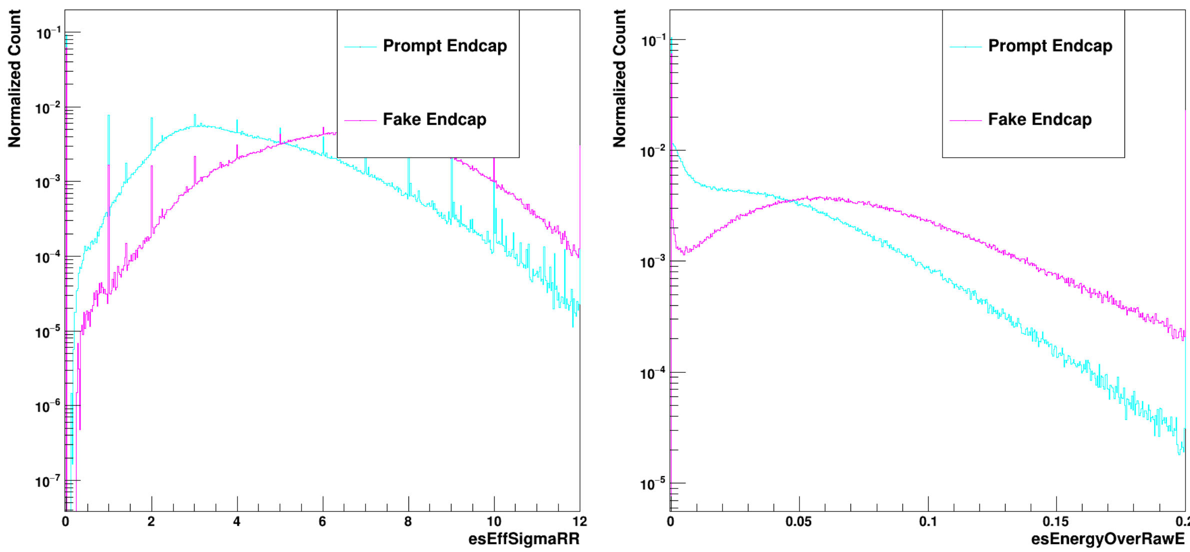



**(a)** Chg. Iso. W.r.t. Worst Vertex

**(b)** Chg. Iso. W.r.t. Chosen Vertex

**Figure 6.6.** Plots of the charged isolations with respect to the chosen and worst vertices. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The final two variables in the model are only used in the endcap models. In fact, these variables only exist for endcap events.




**(a)** ES Eff. Sigma RR

**(b)** ES Energy over Raw E

**Figure 6.7.** Plots of two endcap-specific variables: esEffSigmaRR and esEnergyOverRawE. The prompt signal is shown in light blue, whereas the fake background is shown in pink.

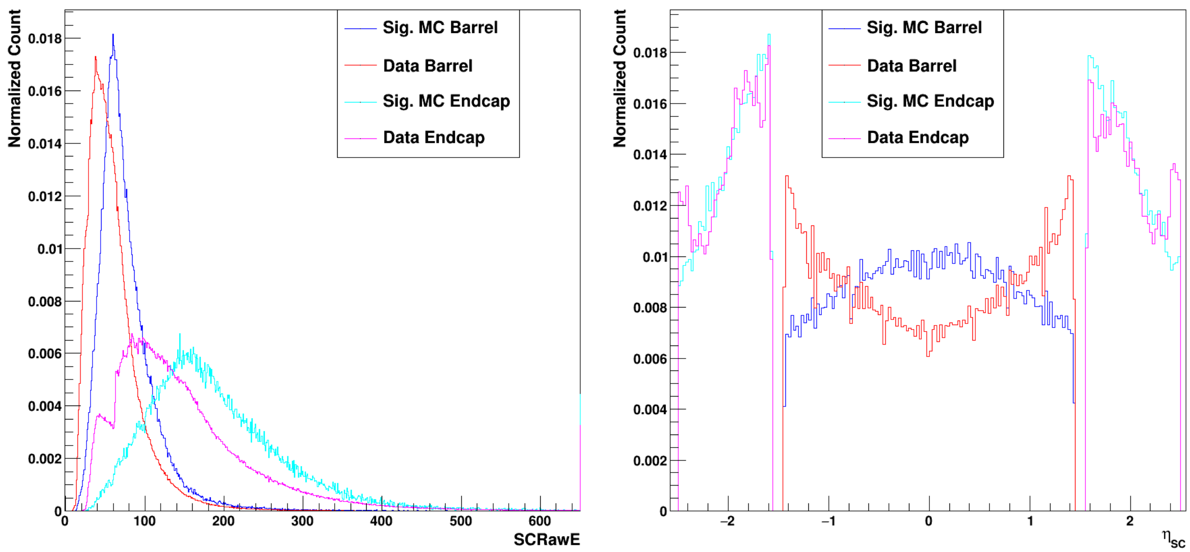
Comparing to the MVA-HLT variables shown in 5.1, it can be seen that a few variables have been added at the offline selection level. The photon isolation is analogous to the ECalIso used at the trigger level, but is calculated differently (how?). The two charged isolations used are entirely new additions, as the MVA-HLT is calculated before any complex vertexing is implemented. (Why are both used?). (Statements on rho, and the endcap variables)

The hyperparameter optimization at the offline level was performed very similarly to the MVA-HLT; the max depth and learning rate were varied in tandem until an optimal point was found. The same process of early stopping within XGBoost was used, as well. The hyperparameters were, however, found to be quite different from the trigger-level optimization. Since there are considerably more input variables used (13/15 for the barrel/endcap versus the 8 for the trigger), the max depth was necessarily higher. Consequently, the learning rate was also made lower. The optimum hyperparameters were, then, found to be a maximum depth of 18 and a learning rate of 0.06.

### A Data/MC BDT for $H \to \gamma\gamma$

While the above $\gamma + Jet$ BDT can help differentiate real photon candidates from fakes, it cannot, in principle, properly differentiate a photon candidate coming from a signal process from a photon candidate coming from a background process. Since the EGamma trigger selections will already cut out many events with jet-like photon candidates, much of the $H \to \gamma\gamma$ preselection is aimed at removing real photons coming from non-resonant processes (or even non-Higgs resonances). For the standard preselection, the $p_T$ and $p_T/M_{\gamma\gamma}$ cuts achieve this.

Furthermore, the $\gamma + Jet$ BDT does not properly mirror the MVA-HLT; since the trigger-level BDT was training using Higgs MC as the signal and real data as the background, the offline selection should employ a similar strategy. So, in the same way that the trigger was trained using signal photon candidates originating from a ggH MC sample and background photon candidates originating from the data. The same input variables used in the $\gamma + Jet$ BDT were then used to train and optimimize the new model. These input variables are plotted in the same way as for the previous section, with the barrel and endcap signal being shown in dark and light blue, respectively. Meanwhile, the barrel and endcap background will be shown in red and pink.

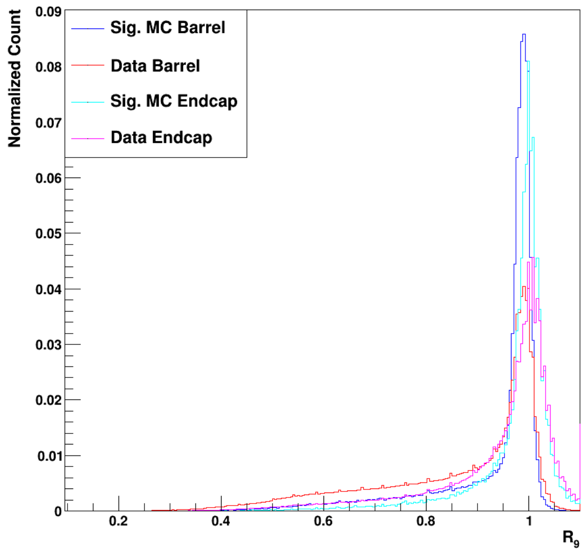



**(a)** Raw Energy

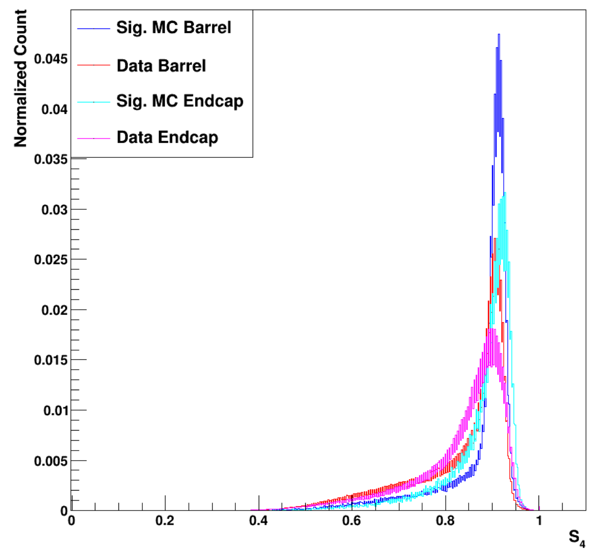
**(b)** $\eta_{SC}$

**Figure 6.8.** Plots of the Supercluster raw energy and $\eta SC$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

Similarly to the $\gamma +$ Jet distributions, a clear difference can be seen in 6.8a between the barrel and endcap distributions of the signal and background. In contrast, however, the difference between signal and background is less pronounced for the data and signal MC. As a result, this will be a less powerful discrimination variable for this new BDT. The features of the $\eta_{SC}$ distributions for the data and MC differ considerably more than those for the $\gamma +$ Jet prompt and fake distributions, however. Particularly in the barrel, it can be seen that the signal MC tends to be much more centeral, whereas the data peaks strongly towards to high-$\eta$ regions of the barrel. The endcap distributions, however, are much more similar than they were for the $\gamma +$ Jet.



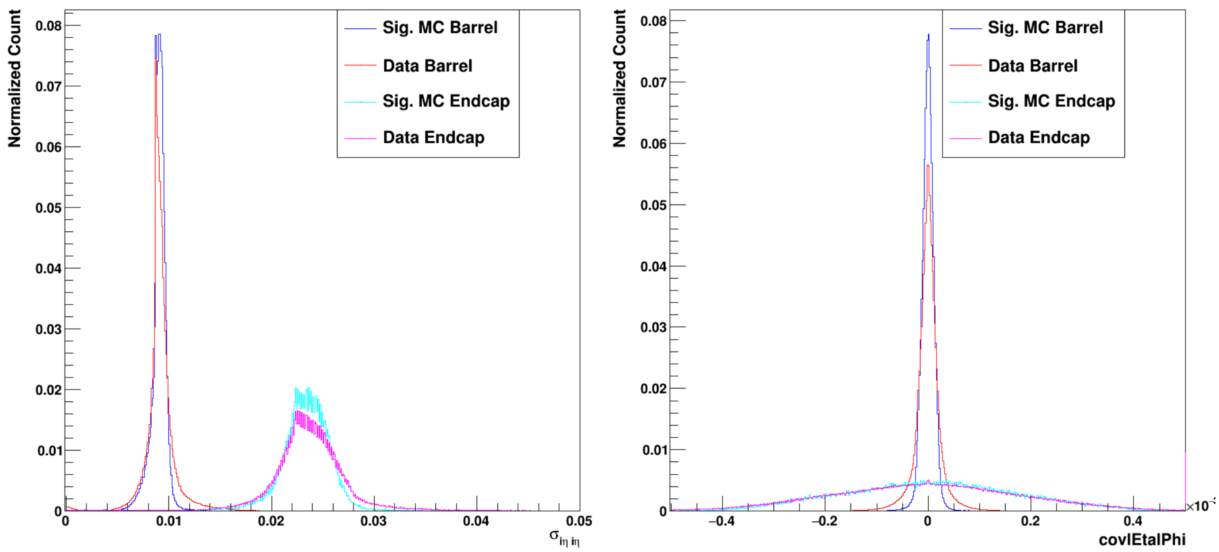

**(a)** $R_9$




**(b)** $S_4$

**Figure 6.9.** Plots of the $R_9$ and $S_4$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

As with the $\gamma +$ Jet distributions, the $R_9$ (figure 6.9a) and $S_4$ (figure 6.9b) variables for the signal tend to be much more strongly peaked near 1 compared to the background.

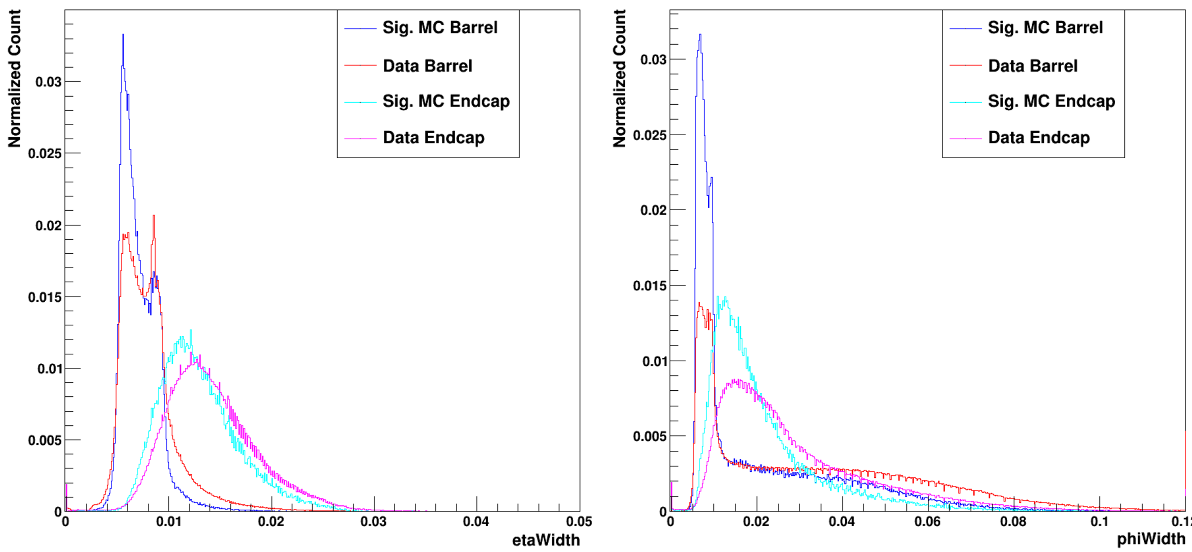



**(a)** $\sigma_{i\eta i\eta}$

**(b)** $\mathrm{cov}(i\eta i\phi)$

**Figure 6.10.** Plots of the $\sigma_{i\eta i\eta}$ and $\mathrm{cov}(i\eta i\phi)$. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

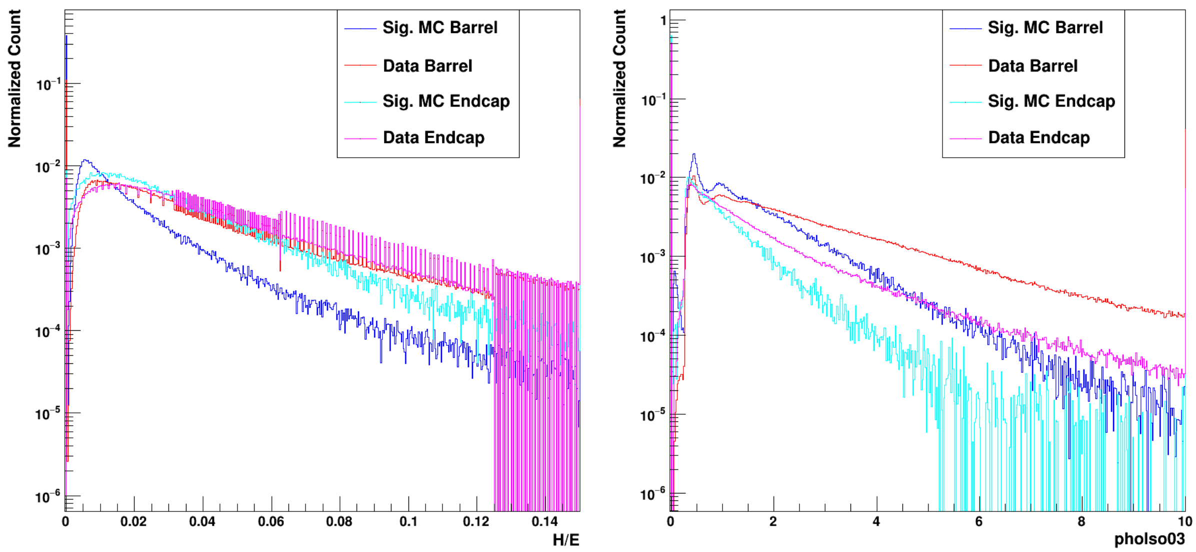
In figure 6.10, it can be seen that the signal and background distributions are quite similar for both $\sigma_{i\eta i\eta}$ and $\mathrm{cov}(i\eta i\phi)$. Compared to the $\gamma+$ Jet BDT, this means that these showershape variables will be much less powerful. Likely, due to the MVA-HLT selection applied, many of the more jet-like photons have already been removed from this training sample.




**(a)** $\eta$ Width **(b)** $\phi$ Width

**Figure 6.11.** Plots of the $\eta$ and $\phi$ widths. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

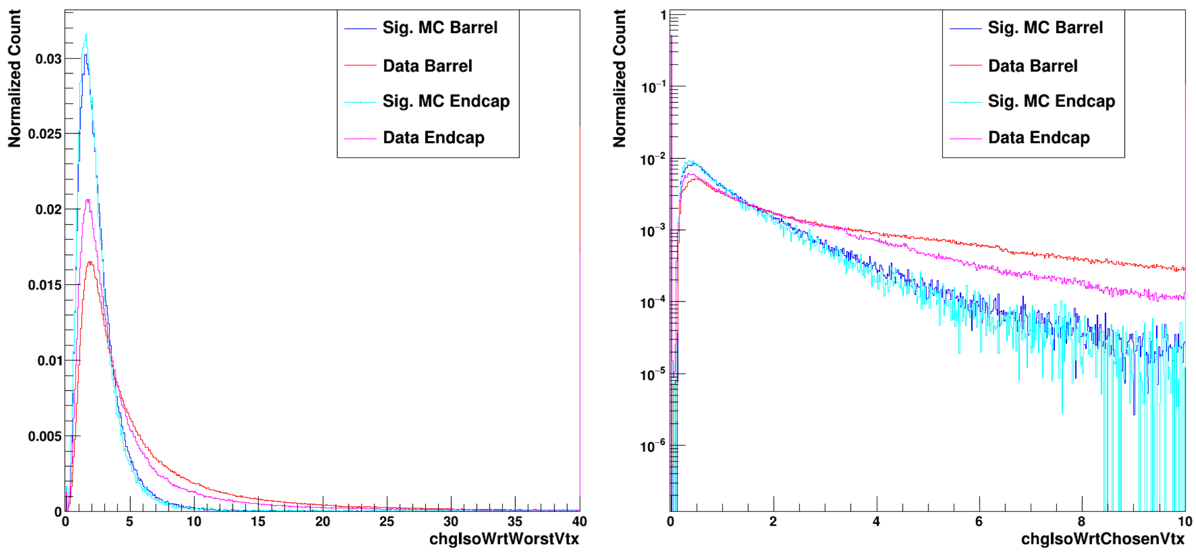
The $\eta$ and $\phi$ widths in figure 6.11 show similar separation between signal and background relative to the $\gamma$ + Jet distributions.




**(a)** H/E **(b)** Photon Isolation

**Figure 6.12.** Plots of the H/E and photon isolation. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The H/E and photon isolation in figure 6.12 show decent separation between the signal MC and data in both the barrel and endcap. The barrel and endcap for the data are quite similar for H/E, while the signal MC shows considerably more variation between the two detector elements. Overall, H/E remains a quite powerful discrimination variable in these new samples. Similarly, clear differences can be seen between the signal and background in the photon isolation.

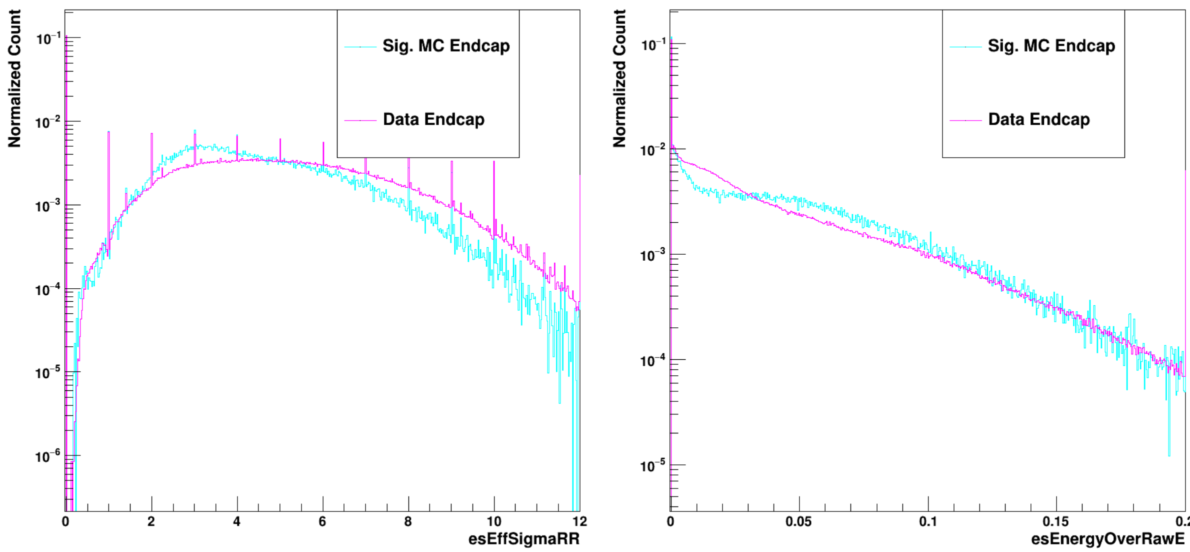



**(a)** Chg. Iso. W.r.t. Worst Vertex

**(b)** Chg. Iso. W.r.t. Chosen Vertex

**Figure 6.13.** Plots of the charged isolations with respect to the chosen and worst vertices. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

In figure 6.13, it can immediately be seen that the two charge isolations are quite similar for the signal and background in both the barrel and endcap. While the values tend to extend a bit higher for the background overall, the shapes are far more similar between the signal and background classes relative to the $\gamma +$ Jet distributions.




**(a)** ES Eff. Sigma RR **(b)** ES Energy over Raw E

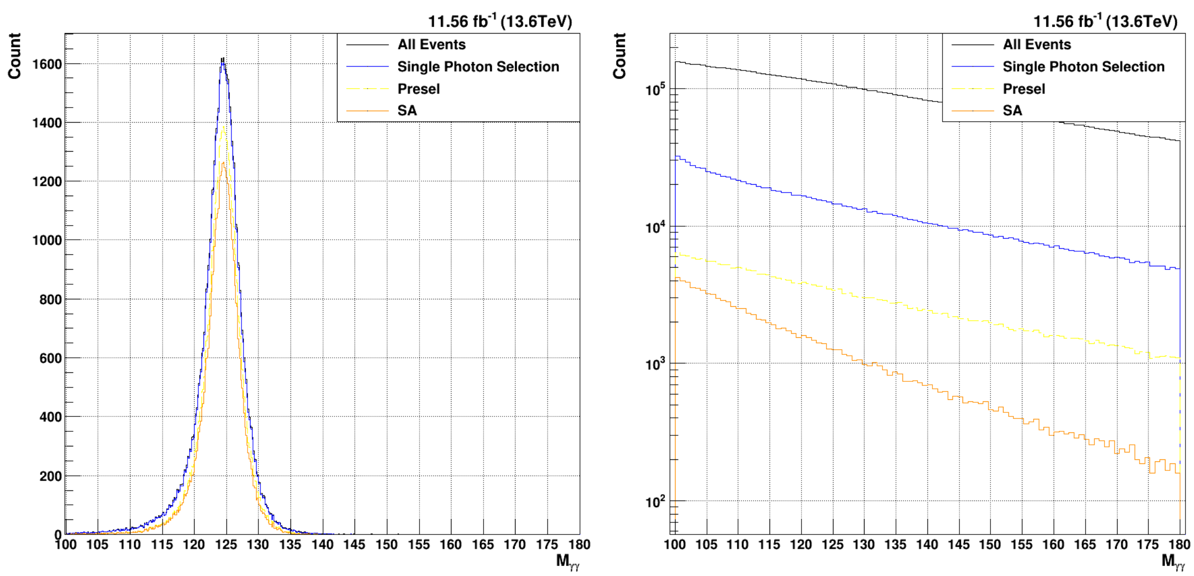
**Figure 6.14.** Plots of two endcap-specific variables: esEffSigmaRR and esEnergyOverRawE. The prompt signal is shown in light blue, whereas the fake background is shown in pink.

The same process of hyperparameter optimzation as the previous models was used. Notably, the signal MC and data show much more similarity in the input variable distributions relative to the $\gamma +$ Jet samples. As a result, the model must become a bit more complex to achieve the desired discrimination power. The same experimental optimization process led to a maximum depth of 18 paired with a learning rate of 0.025. While the maximum depth is the same as that of the $\gamma +$ Jet model, the learning rate is quite a bit lower than the 0.06 used for that BDT.

### 6.2.2 Determination of Loose Single Photon MVA Cuts for $H \to \gamma\gamma$

The preexisting preselection first makes decently loose cuts on individual photons, then tighter $p_T$ and $p_T/M_{\gamma\gamma}$ cuts on selected diphoton combinations. A very similar strategy can be used for the BDT-based offline selection. Firstly, loose cuts on the above BDT scores can be applied to the individual photon candidates, which will reduce the number of possible diphoton combinations and cut out many events with no reasonable photon candidates. Then, with the list of individual photons cleaned up a bit, a tighter selection can be applied to the resulting diphoton objects for the selection of a final pair of photons.

So, then, the aim is to choose a loose set of cuts on the $\gamma + Jet$ and Data/MC BDT scores before applying a tighter final selection on the resulting diphoton combinations. One great advantage of a BDT-based approach is the ease with which these cuts can be adjusted and optimized; while a many-variable preselection has several different cuts to optimize, a binary classifier BDT only has a single output score on which to make cuts. While the goal of this current offline selection is to demonstrate an example analysis that leverages the increased signal efficiency of the MVA-HLT, the existence of these single photon scores allows very easy tuning of the photon preselection for any given analysis. The example cuts chosen below were designed to reduce the signal efficiency by about 2%, which primarily serves to reduce the number of possible photon combinations for final selection; ultimately, the diphoton cuts described in 6.2.3 will be the primary final selection used to achieve the desired data rate.

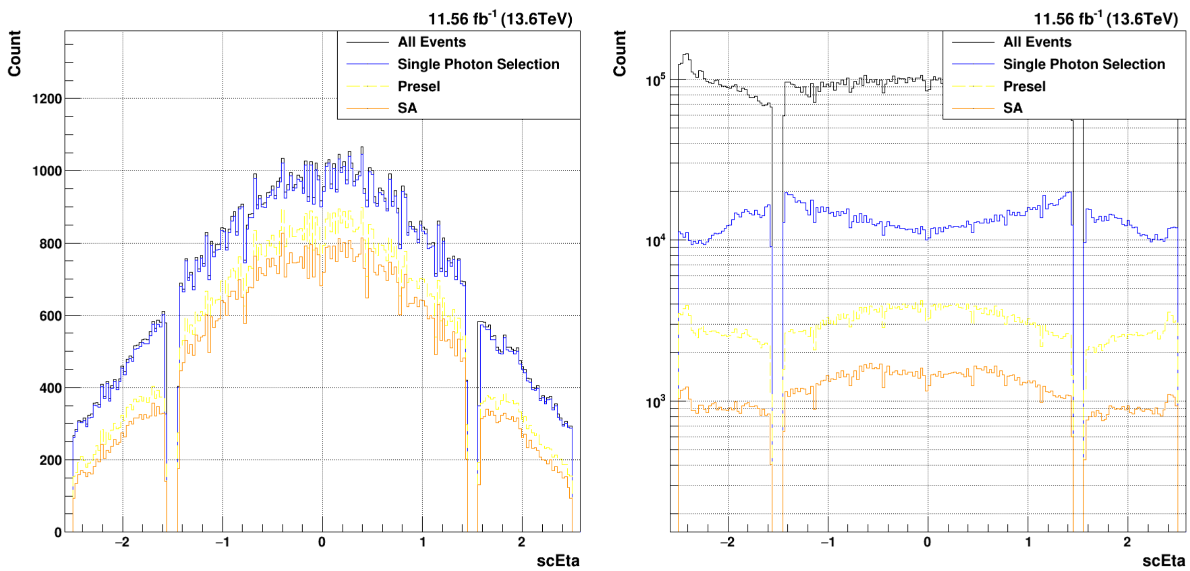



**(a)** Signal MC $M_{\gamma\gamma}$ **(b)** Data $M_{\gamma\gamma}$

**Figure 6.15.** Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$

Clearly, as seen in figure 6.15, these chosen single photon cuts (in blue) leave the signal MC relatively untouched. The final chosen signal efficiency for this gluon fusion $H \to \gamma\gamma$ is

98.3%, which is much higher than the standard single photon preselection efficiency of 78.5%. Of course, it can be seen that the background rate is much higher – while the standard preselection cuts achieve a background rejection of 96.8%, the loose single photon BDT cuts only achieve an 85.9% rejection.

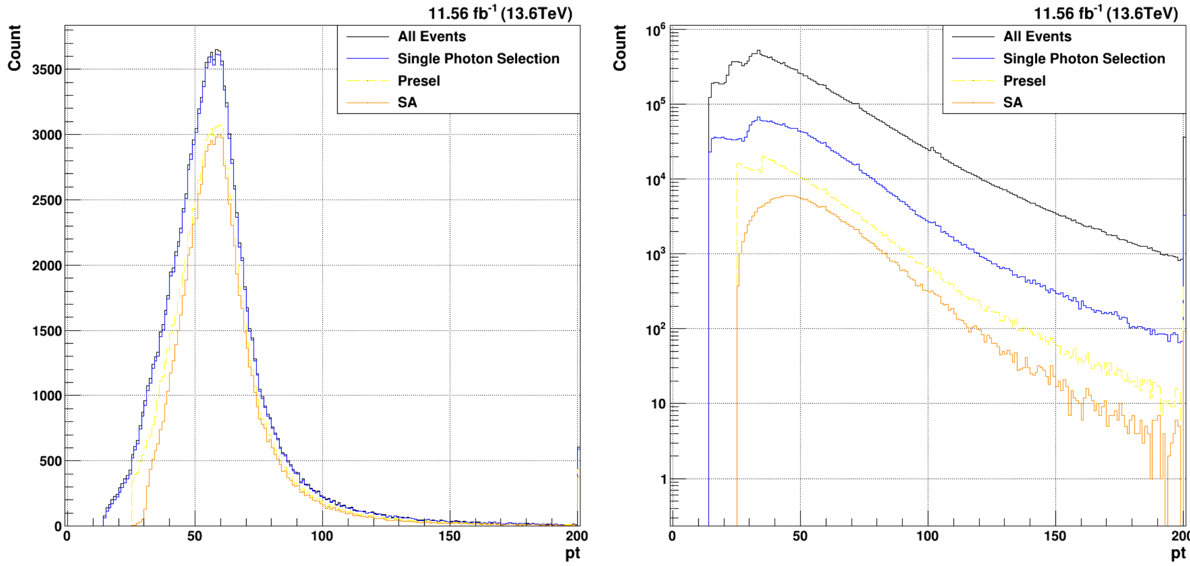



**(a)** Signal MC $\eta_{SC}$

**(b)** Data $\eta_{SC}$

**Figure 6.16.** Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $\eta_{SC}$. Both the lead and sublead photons are plotted together.

Maximizing the kinematic acceptance of the signal events in $\eta_{SC}$ is a key consideration to any photon selection. In figure 6.16, the acceptance of the different single photon selections is shown. Given that the signal acceptance is so high for these loose single photon cuts, it can be seen immediately that there are no regions of the detector shown for which the acceptance is problematic. Considering the preselection (yellow), however, a relative drop in efficiency can be seen in the low-$\eta$ regions at the start of the endcap (near $|\eta| = 1.5$). Investigations showed that the hard $R_9$ cuts placed on the photons by the standard preselection are responsible for this relative dip in efficiency. Utilizing the $R_9$ in the BDT, as opposed to these hard cuts, ensures that the kinematic coverage for the loose single photon BDT selection is better than that of the standard preselection.




**(a)** Signal MC $p_T$ **(b)** Data $p_T$

**Figure 6.17.** Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $p_T$. Both the lead and sublead photons are plotted together.

A major advantage to these loose single photon cuts can be seen in figure 6.17. Clearly, the standard preselection cuts considerably reduce the signal MC kinematic phase space by making relatively tight cuts of $p_T^{Lead} > 35GeV/c^2$ and $p_T^{Sub} > 25GeV/c^2$. While this also leads to a considerable reduction in data rate, the loss of good Higgs events with low-$p_T$ photons may impact future analyses with different kinematics. By comparison, the loose single photon cuts, being based entirely on the single photon BDT scores, do not noticably impact the $p_T$ distributions of either the signal or background.

### 6.2.3 Creation of Diphoton BDT for $H \to \gamma\gamma$

With the single photon preselection cuts determined, an additional selection must be placed on the diphoton objects. The most obvious continuation of the current strategy is to create an additional MVA that now works on diphoton quantities. Previous analysis strategies have, in fact, employed a diphoton MVA for the classification of events [12]. A new diphoton BDT can be constructed in reference to this, with the aim of making final cuts on this score such that the

**Table 6.3.** Table of diphoton BDT input variable names, and a short description of each.

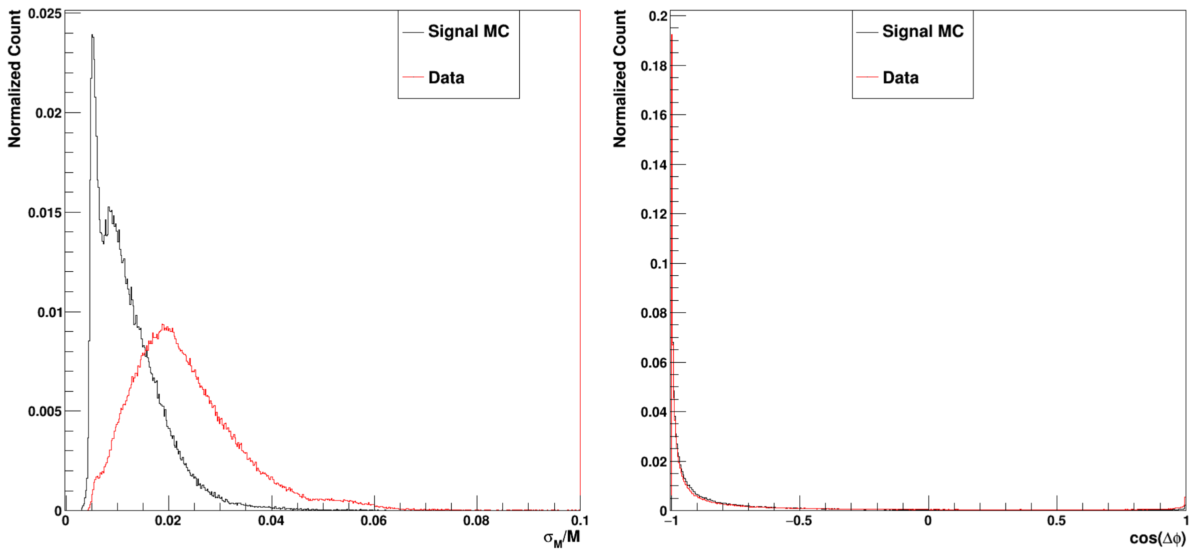
| Variable Name | Description |
|---|---|
| $p_T^{Lead}/M_{\gamma\gamma}$ and $p_T^{Sub}/M_{\gamma\gamma}$ | The mass-normalized $p_T$ for each photon |
| $\eta_{sc}^{Lead}$ and $\eta_{sc}^{Sub}$ | The supercluster $\eta$ for each photon |
| $BDT_{\gamma+jet}^{Lead}$ and $BDT_{\gamma+jet}^{Sub}$ | The $\gamma+jet$ BDT score for each photon |
| $BDT_{Data/MC}^{Lead}$ and $BDT_{Data/MC}^{Sub}$ | The Data/MC BDT score for each photon |
| Lead and Sublead Electron Veto | A boolean to reject photons origininating from electrons |
| $cos(\Delta\phi)$ | The cosine of the $\phi$ separation of the two photons |
| $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ | A relative mass resolution estimate |

background rate is controlled while maintaining maximum signal efficiency.

While the goal of the single photon BDTs is to reject individual candidates that appear to not be photons, the diphoton BDT aims to properly differentiate diphton candidates coming from $H \to \gamma\gamma$ from both events that do not contain two real photons and events with two real photons originating from background processes. As a result, the inputs to this model will consist primarily of lead and sublead single photon variables. By providing both the lead and sublead values to one model, correlations can be found that ensure that selected events contain two real photons with the necessary kinematics to possibly comprise a signal event. Below, in table 6.3, each of the input variables is listed and described. These were largely adopted from the Run2 Diphoton photonID, with a few differences.

Since the goal of this model is to differentiate signal diphoton combinations from those in the background, signal MC and data can again be used for the signal and background for the BDT, as with the data/MC BDT described above. Again, the input variable distributions can be examined to see the separation between signal and background class events.

The first two variables are related directly to the kinematics of the diphoton object, and therefore are not split into lead and sublead values. The first, $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$, provides a relative invariant mass resolution estimate of the diphoton combination. It can clearly be seen in figure 6.18a, that there is great separation between the signal MC and data events. The signal MC tends to have a much lower $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$, which corresponds to a higher mass resolution. This is to be expected, as the two selected photons in the $H \to \gamma\gamma$ are much more likely to come from a resonant process than those in the data. The $\cos(\Delta\phi)$, however, does not exhibit the same strong separation. This variable aims to measure the separation in $\phi$ of the two selected photon candidates. Based on the other selections at play, however, it seems that this variable will not be a strong discrimination variable.

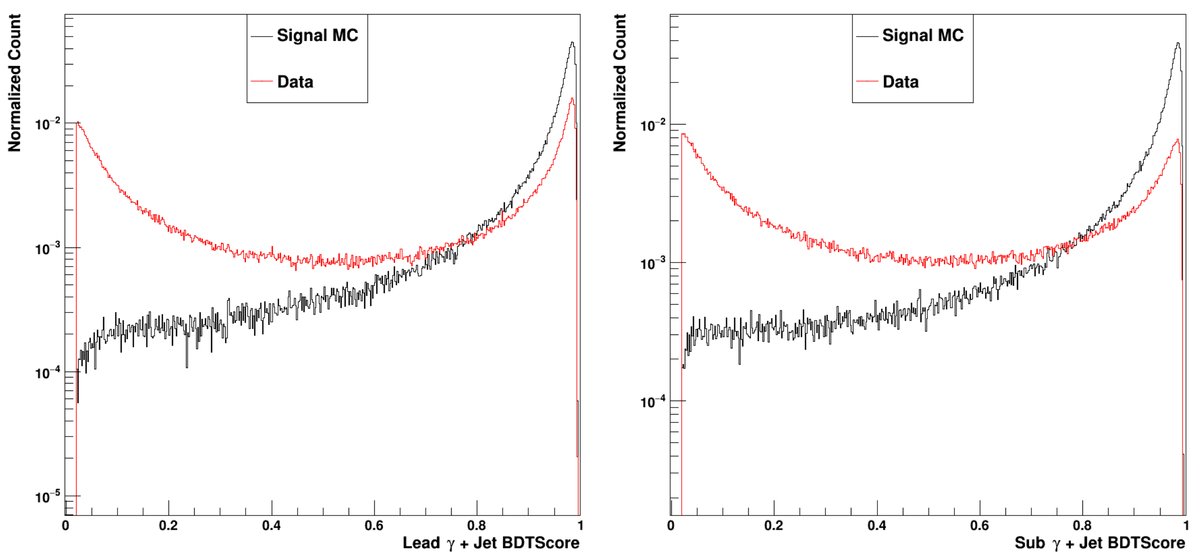



**(a)** $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ **(b)** $\cos(\Delta\phi)$

**Figure 6.18.** The two per-event diphoton input variables,$\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ and $\cos(\Delta\phi)$. The signal MC is in black, whereas the data events are in red.

The next two pairs of input variables are the single photon BDT scores. Firstly, the $\gamma+$ Jet single photon BDT scores are shown below in figure 6.19. As with all XGBoost BDT binary classification scores, a value near 0 shows that the algorithm has identified the photon as being part of the background class, while a score near 1 corresponds to an event identified as part of the signal class. For the signal MC, it can be seen that the scores peak very well near 1. This is to be expected, as the signal MC photons here are matched to a generated photon and therefore guaranteed to be a real photon. This effect is clear for both the lead and sublead photons.

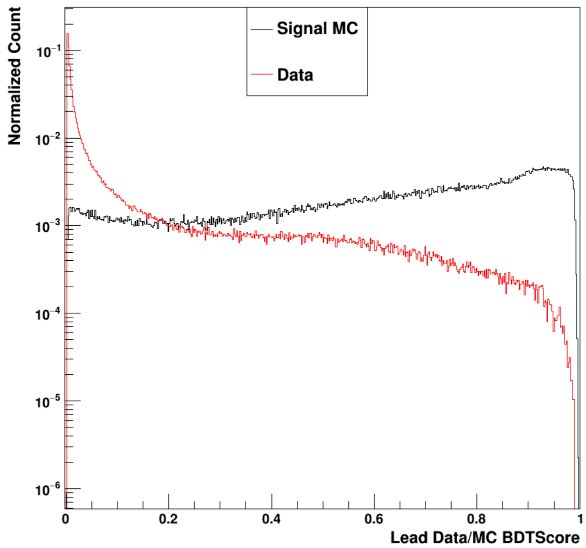
Clearly, however, the background photons in red are not constrained entirely near 0. In fact, there are two peaks in both the lead and sublead score distributions. While many of the background photon candidates can easily be assigned a score near 0, there are also a large number for which the $\gamma +$ Jet scores are not adequate to differentiate the background from the signal. This likely due, at least in part, to the composition of the data overall. While many of the data events consist of dijet QCD background, there are likely many events for which one or even two real photons have been selected. This is exacerbated by the triggers required for this training sample; here, since either the MVA-HLT trigger or the standard diphoton trigger must be passed to enter the training sample, the odds of a data event containing one or two real photons is much higher than it would be for other data samples. This demonstrate why the $\gamma +$ Jet BDT alone is not adequate to correctly differentiate between signal and background events.



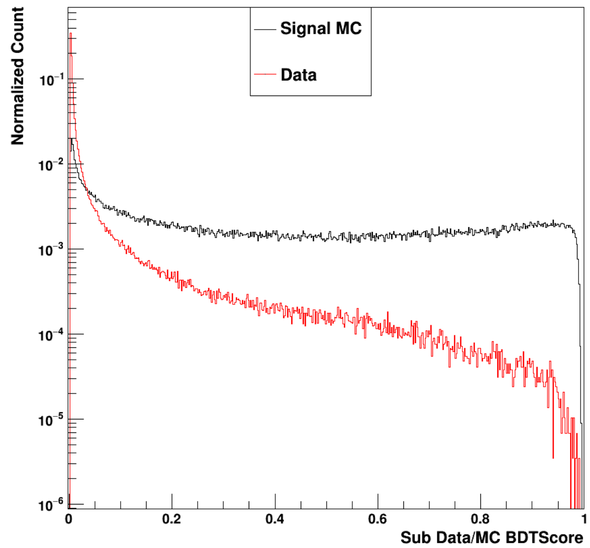

**(a)** Lead $\gamma +$ Jet BDT Score **(b)** Sub $\gamma +$ Jet BDT Score

**Figure 6.19.** The lead and sublead $\gamma +$ Jet XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

Secondly, the data/MC scores for both the lead and sublead photons are shown in figure 6.20 on a logarithmic scale. As expected, there is clear separation between the signal MC in black and the data in red. It is expected that the signal MC distributions will largely cluster near 1, which corresponds to a positive identification as signal. While this is true for both the lead and sublead scores, the identification of the background is perhaps a stronger effect. It can be seen




**(a)** Lead Data/MC BDT Score



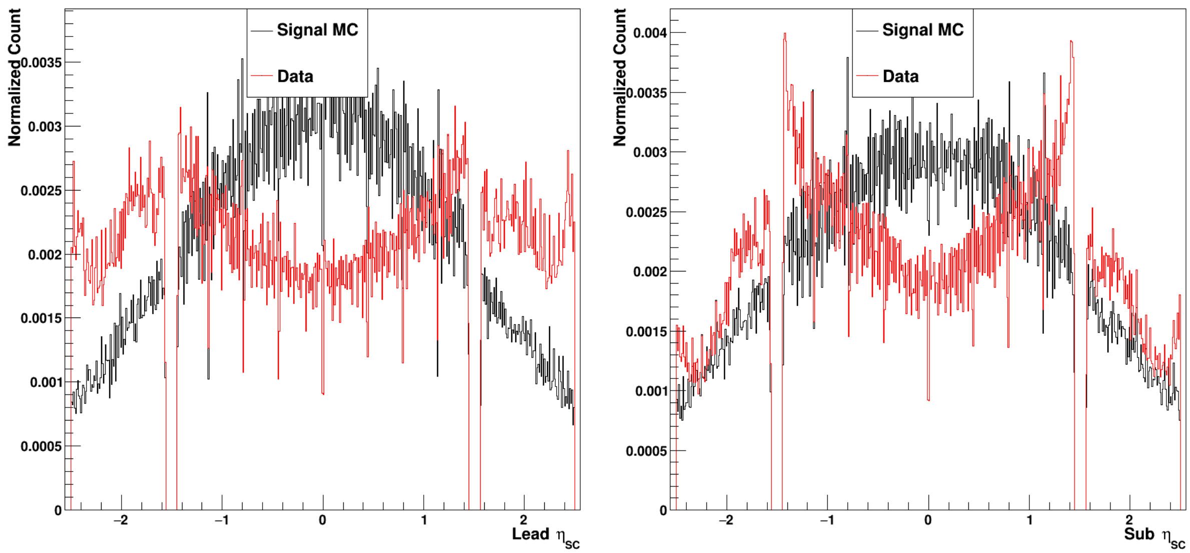

**(b)** Sub Data/MC BDT Score

**Figure 6.20.** The lead and sublead Data/MC XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

that the data events are very closely clustered near 0. Very few of the background events, on the other hand, are placed near a score of 1.

Similarly to the $\gamma$ + Jet single photon BDT, then, it is clear that this data/MC BDT is not adequate alone to differentiate signal and background. The diphoton BDT, then, is a crucial step of the signal and background selection; by having access to both the single photon BDTs, the diphoton BDT can construct correlations that allow the final model to be considerably more discerning than input scores it uses. As will be shown in chapter 7, this works to great effect.

While the single photon quality scores given by the single photon BDTs are quite helpful, the diphoton kinematics are additionally a very strong source of discrimination power. As sean for both the lead (figure 6.21a) and sublead (figure 6.21a) $\eta_{SC}$ distributions, the shape of the single Higgs signal MC in black and the data in red are quite different. Due to the kinematics of the Higgs production, as discussed in 1, the signal MC tends to be much more central in the barrel detector (closer to perpendicular to the beamline). Meanwhile, in the endcaps, the amount of signal photons rapidly drops as the edges of the chose $\eta_{SC}$ range are approached. Due to this good separation between signal and background, the $\eta_{SC}$ of the lead and sublead photon provide good discrimination power to the binary classifier BDT.

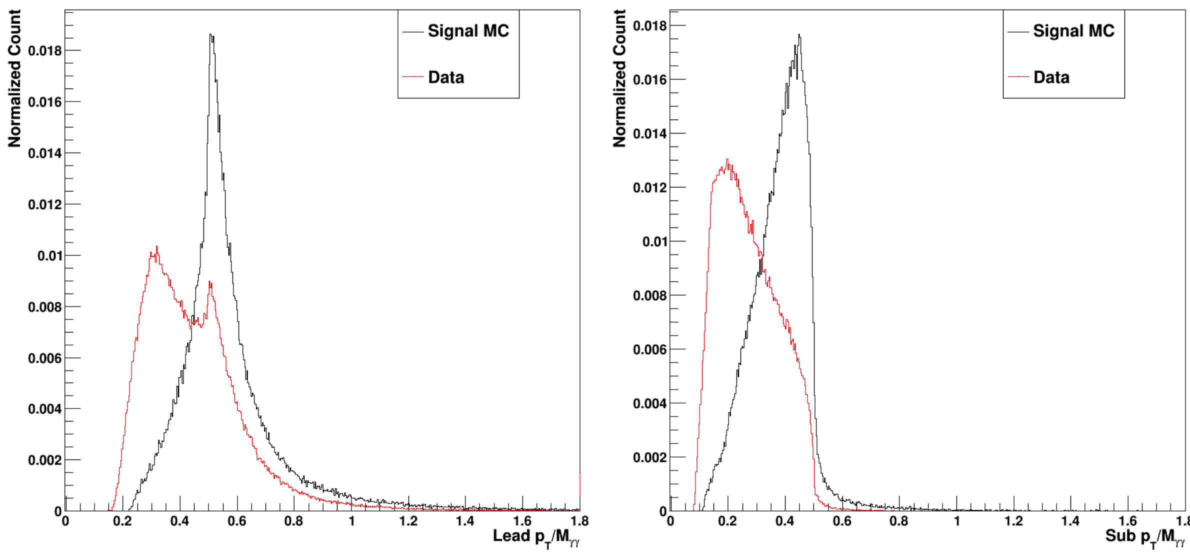



**(a)** Lead $\eta_{SC}$

**(b)** Sub $\eta_{SC}$

**Figure 6.21.** The lead and sublead $\eta_{SC}$. The signal MC is in black, whereas the data events are in red.

The other diphoton kinematic quantities used by the diphoton BDT are the $p_T/M_{\gamma\gamma}$ of the two selected photons. This mass-normalized measure of the photon candidates' transverse momentum is used in the standard diphoton preselection given its great discrimination power. Of course, if hard cuts on a variable are successful in separating signal from background, a BDT should be able to leverage the information to even greater effect. Furthermore, as with the $p_T$ cuts, removing the hard cuts on this variable considerably improve the kinematic coverage of the offline selection. In figure 6.22, a clear separation between the signal and background events can be seen for both the lead and sublead photon candidates. Paired with the corresponding $\eta_{SC}$ values, this lends considerable kinematic discrimination power to the diphoton BDT.

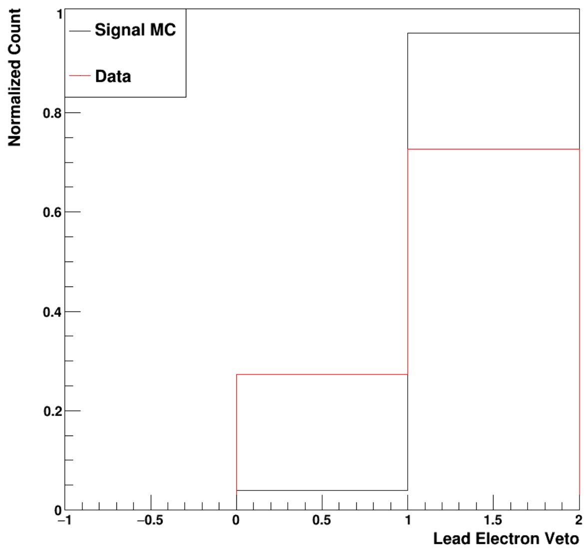



**(a)** Lead $p_T/M_{\gamma\gamma}$ **(b)** Sub $p_T/M_{\gamma\gamma}$

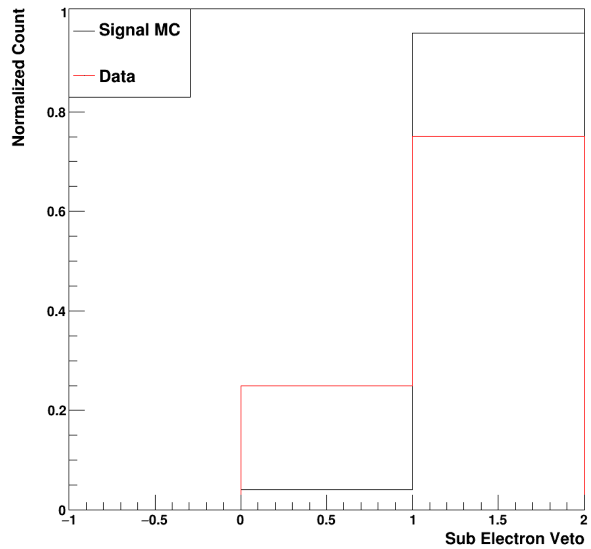
**Figure 6.22.** The lead and sublead $p_T/M_{\gamma\gamma}$. The signal MC is in black, whereas the data events are in red.

Finally, the electron veto values can be used to help differentiate between photons and electrons in the final selection. As discussed in section 3.4.1, photons and electrons are quite hard to differentiate purely from ECal information. The electron veto was designed to leverage charged track information to provide a conversion-safe way to reject electrons from photon candidates [13]. This quantity has not historically been deployed in the diphotonMVA; since the standard preselection generally makes a hard requirement on the electron veto, there is no need to use it in the diphotonMVA. By adding the lead and sublead electron veto decision to the diphoton BDT, however, the information can be correlated with the other variables provided to more elegantly reject electrons in the data while still maximizing the signal efficiency.

The values of the electron veto are shown in figure 6.23. Since this is just a veto decision, the variable is saved as a boolean. As a result, the plots show a value of 0 for false (failing the electron veto) and 1 for true (passing the electron veto). A very clear separation can be seen between the signal MC (black) and data (red). Very few of the selected signal MC lead or sublead photon candidates fail the electron veto, wheareas there are a considerable portion of the data events that fail. It can be seen, however, that there are still many data events that pass the electron veto.




(a) Lead Electron Veto



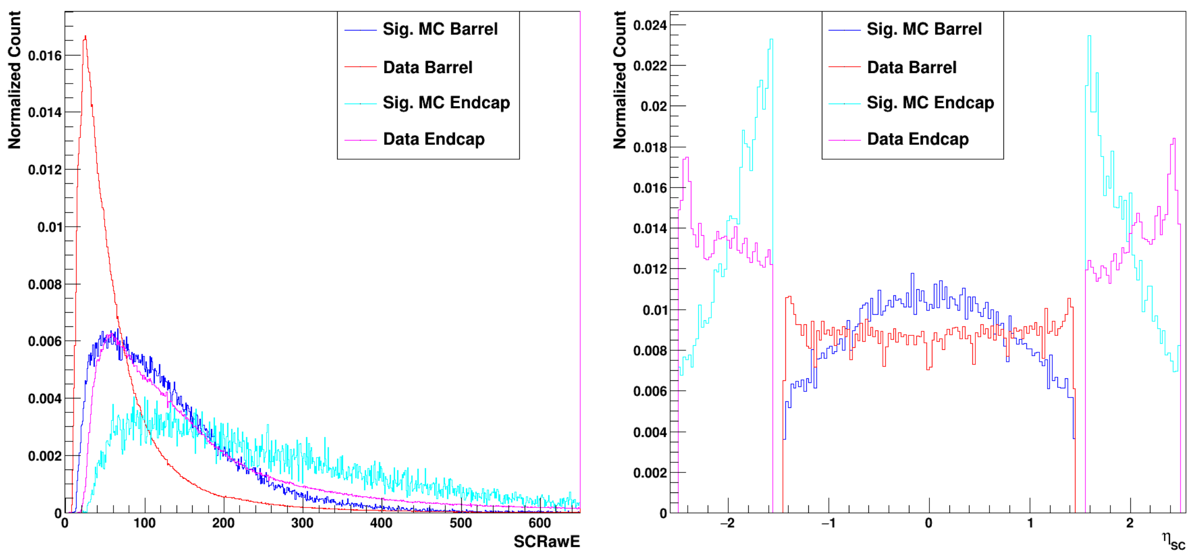

(b) Sub Electron Veto

**Figure 6.23.** The lead and sublead electron veto. The signal MC is in black, whereas the data events are in red.

As with the other two BDTs, the hyperparameters must of course be optimized once the input variables are selected. This was, again, a relatively experimental process. Given the reduced number of input variables relative to the single photon BDTs, lower maximum depths were explored. A max depth of 10 was selected in conjunction with a learning rate of 0.1. This lower max depth and higher learning rate allows to model to still leverage the reduced number of input variables without considerable overtraining.

**A Data/MC BDT for $HH \to b\bar{b}\gamma\gamma$**

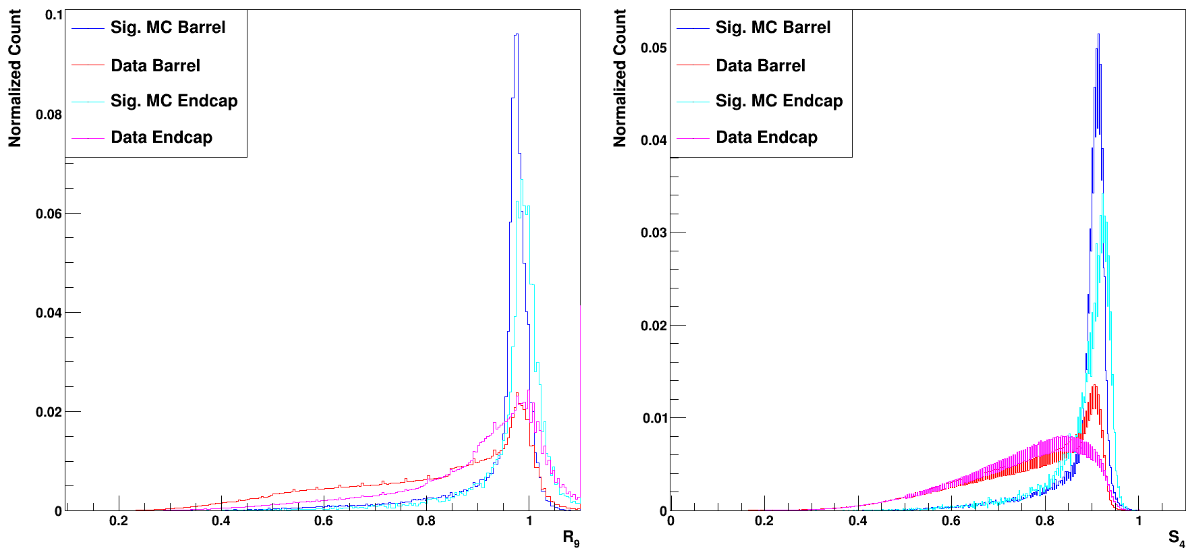
Of course, since the Data/MC and diphoton BDTs are trained using signal MC and data, they should be retrained for samples with markedly different input features. As discussed in chapter 1, the kinematics of the $HH \to b\bar{b}\gamma\gamma$ differ considerably from the $H \to \gamma\gamma$. Therefore, the same process of training and evaluation must be repeated for these new samples. While the same data samples are used for this updated analysis, the $HH \to b\bar{b}\gamma\gamma$ workflow applies additional cuts to the $b\bar{b}$ object as discussed in section 3.4.4. The same input variable distributions for the Data/MC BDT are displayed in figures 6.24-6.30 for these new samples. Major differences between the single and DiHiggs samples will be noted.




**(a)** Raw Energy **(b)** $\eta_{SC}$

**Figure 6.24.** Plots of the Supercluster raw energy and $\eta SC$. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

A major difference in $\eta_{SC}$ exists relative to the single H, as seen in figure 6.24. While the signal MC is relatively similar in the barrel, the endcap is far more concentrated at lower-$\eta$ values. The data, on the other hand, is considerably more flat in both the endcap and barrel due to the changed kinematics based on the requirements placed on finding a $b\bar{b}$ pair. On the whole, this leads the $\eta_{SC}$ being used more strongly by the DiHiggs Data/MC BDT. Since less of the MC is in the extreme regions of the endcap, it will be shown that the selection is biased away from selecting data in these regions.

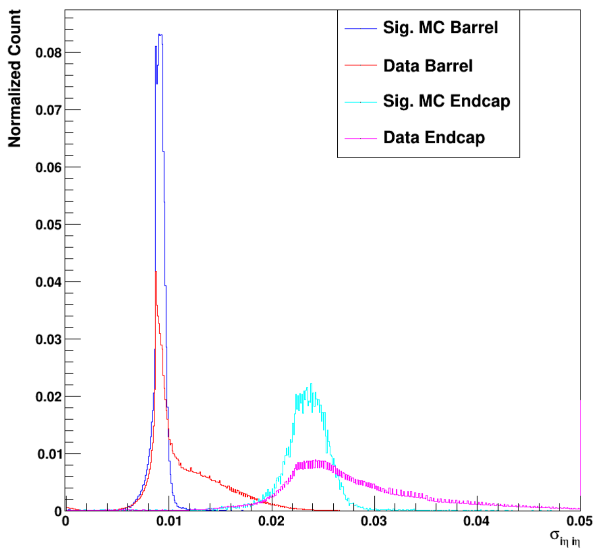



**(a)** $R_9$

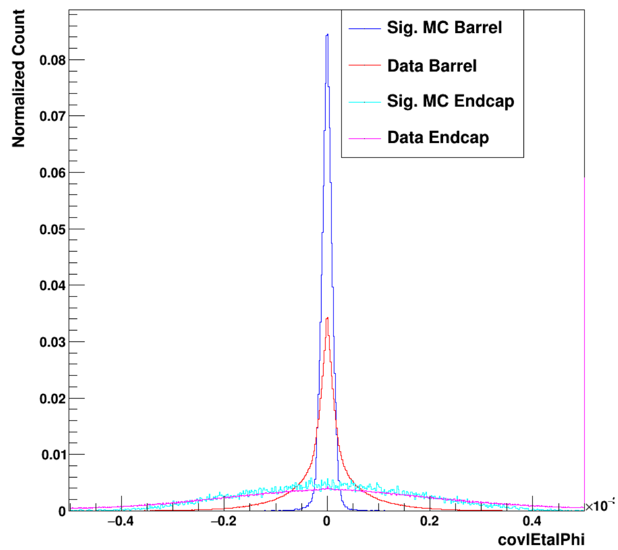
**(b)** $S_4$

**Figure 6.25.** Plots of the $R_9$ and $S_4$. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

Relative to the data selected by the single Higgs workflow, the DiHiggs data selection has considerably lower values on average of both the $R_9$ and $S_4$. This leads to a clearer separation between the signal and background for both the barrel and endcap events.

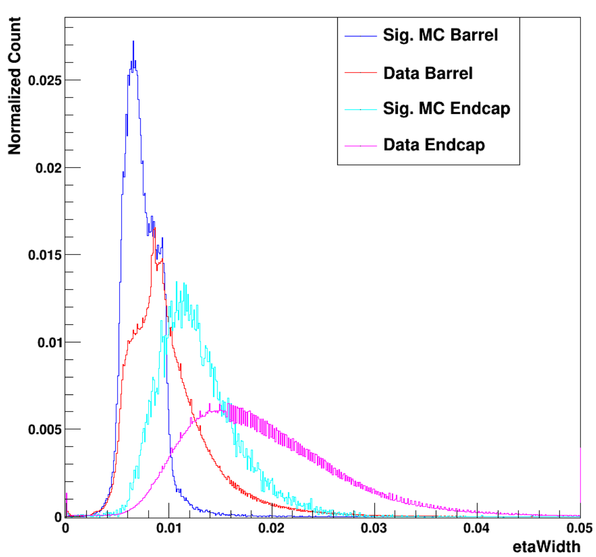



**(a)** $\sigma_{i\eta i\eta}$

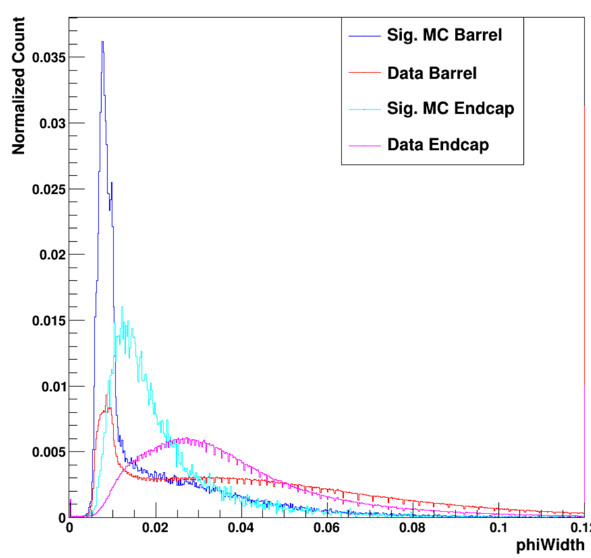



**(b)** cov($i\eta i\phi$)

**Figure 6.26.** Plots of the $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$). The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

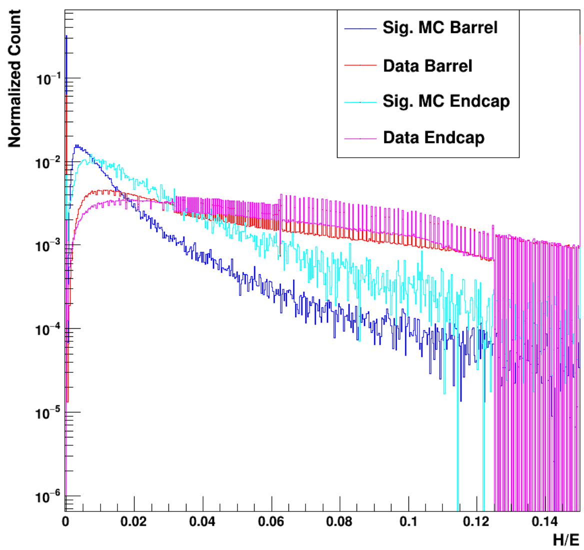



**(a)** $\eta$ Width

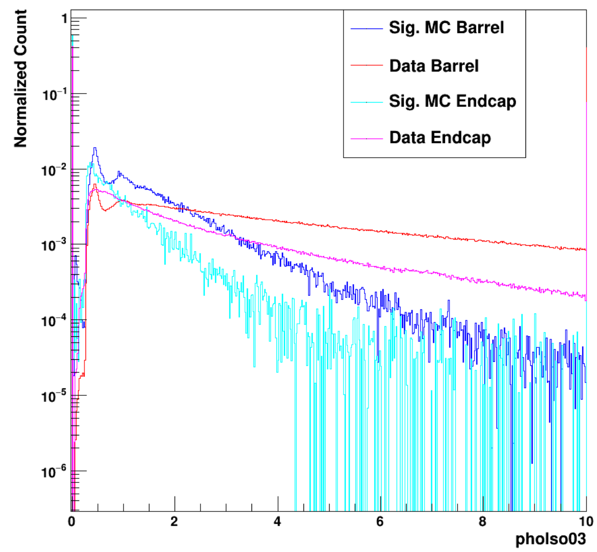



**(b)** $\phi$ Width

**Figure 6.27.** Plots of the $\eta$ and $\phi$ widths. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

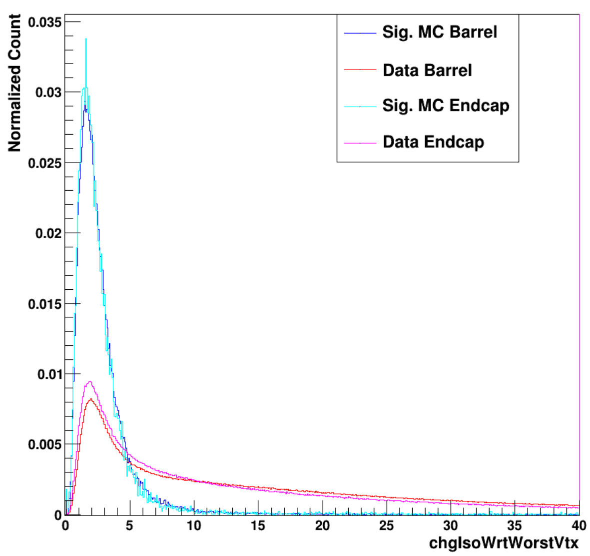



**(a)** H/E

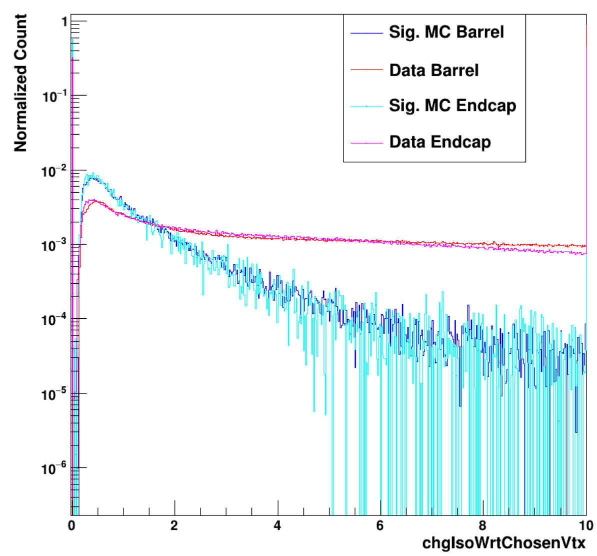



**(b)** Photon Isolation

**Figure 6.28.** Plots of the H/E and photon isolation. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

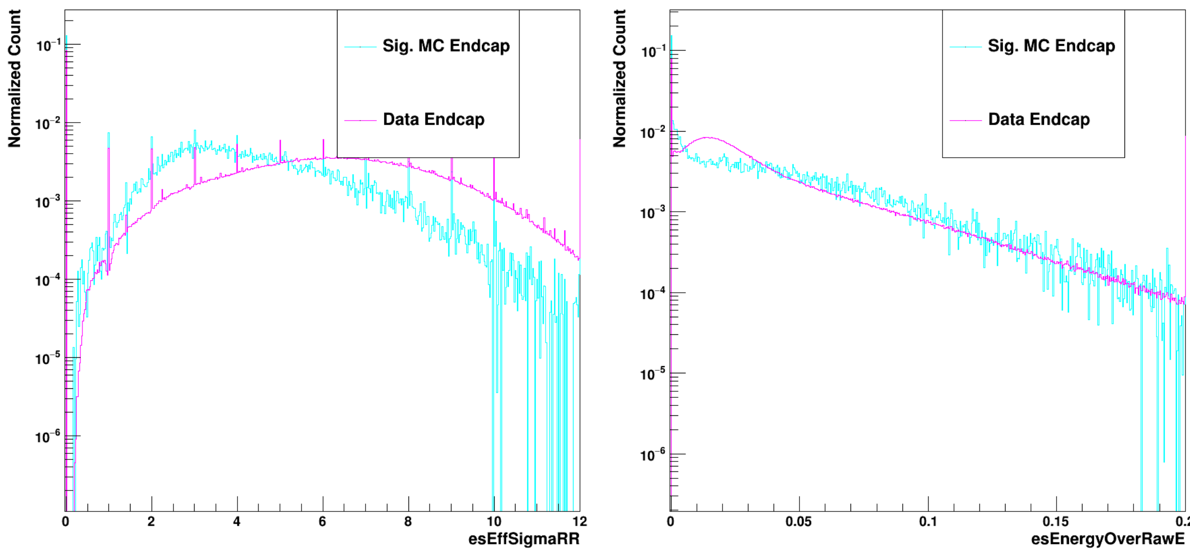



**(a)** Chg. Iso. W.r.t. Worst Vertex




**(b)** Chg. Iso. W.r.t. Chosen Vertex

**Figure 6.29.** Plots of the charged isolations with respect to the worst and chosen vertices. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

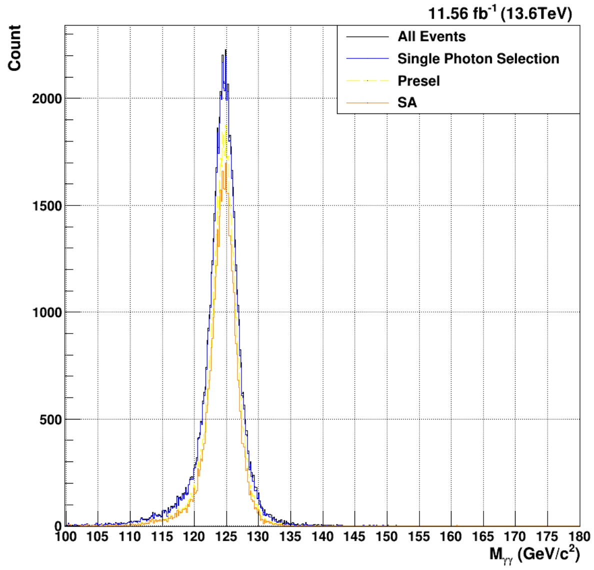



**(a)** ES Eff. Sigma RR **(b)** ES Energy over Raw E

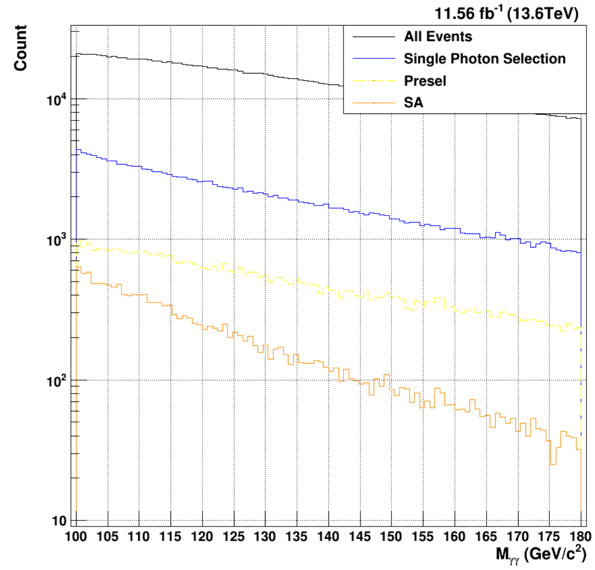
**Figure 6.30.** Plots of the ES Eff. Sigma RR and ES Energy over Raw E. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

### 6.2.4 Determination of Loose Single Photon MVA Cuts for $HH \to b\bar{b}\gamma\gamma$

Given the difference in the resulting output BDT score distributions, the loose single photon cuts must be reexamined for the DiHiggs sample. As previously mentioned, this readjustment of the cuts is a relatively straightforward process. Again, the general goal was to achieve an approximately 98% signal efficiency. Relative to the single H cuts, both the $\gamma +$ Jet and Data/MC cuts are numerically tighter. The chosen cuts yielded a total signal efficiency of 97.35% and a corresponding data acceptance of 14.7%. While both of these values are fractionally lower than that of the single Higgs, there is still ample signal efficiency with which to realize the gains from the trigger level. On the other hand, the preselection yields a signal efficiency of 67.78%, which is considerably lower than that of the single Higgs.

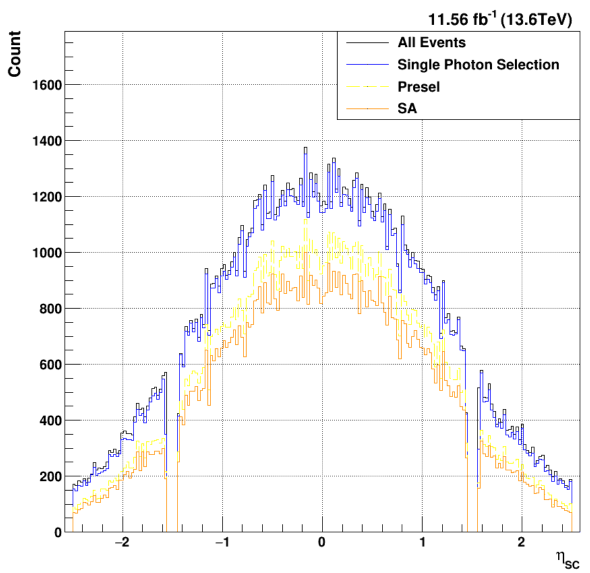



**(a)** Signal MC $M_{\gamma\gamma}$

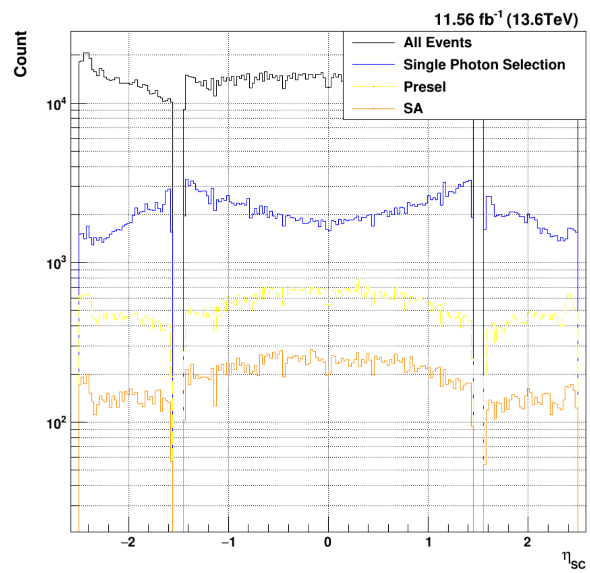



**(b)** Data $M_{\gamma\gamma}$

**Figure 6.31.** Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$




**(a)** Signal MC $\eta_{SC}$

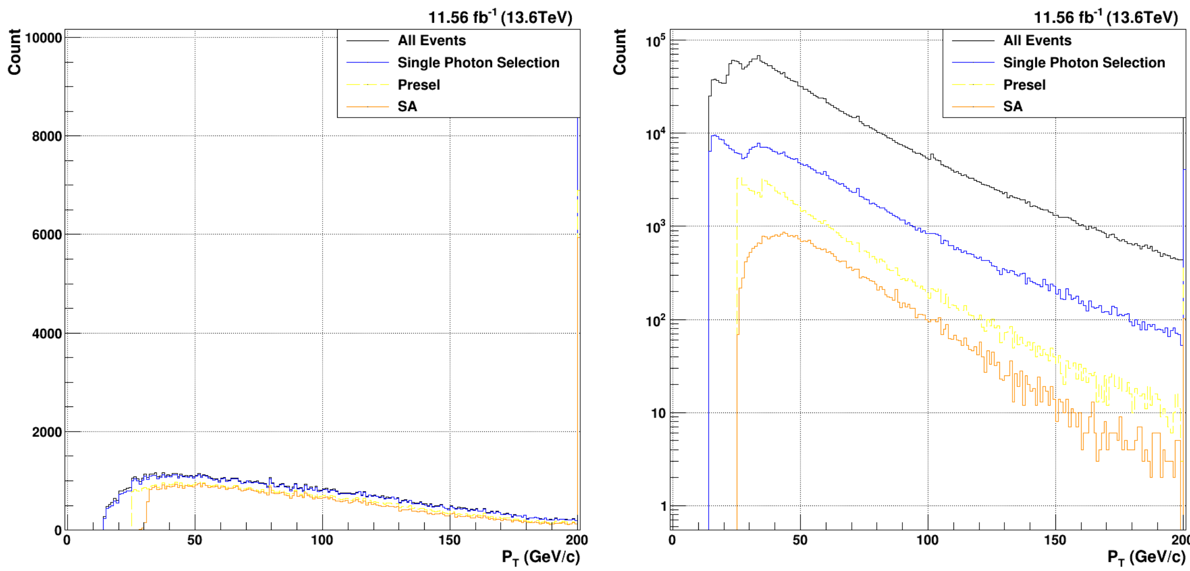



**(b)** Data $\eta_{SC}$

**Figure 6.32.** Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $\eta_{SC}$

Relative to the singleH selection, the DiHiggs data acceptance is relatively unchanged.

Of course, given the very high signal efficiency of this selection, there is little effect on the signal acceptance. A relevant kinematic difference can be seen, however; the DiHiggs signal MC tends to be far more central in the barrel, with very little extending into the endcap. This highlights, again, why retraining both the single photon Data/MC BDT and the diphoton BDT is so crucial for this sample.



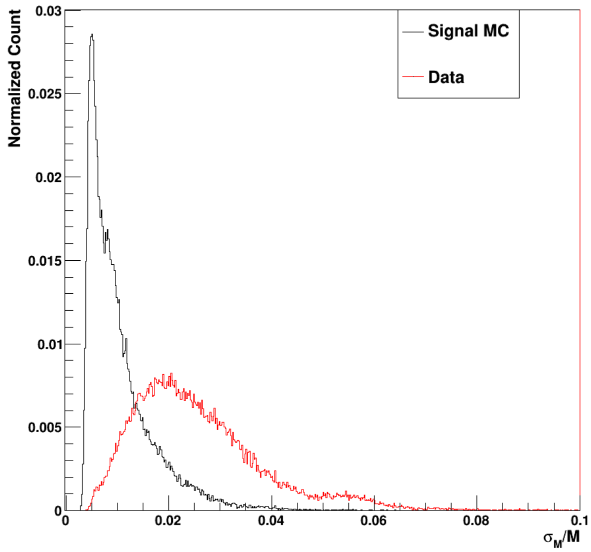

**(a)** Signal MC $p_T$ **(b)** Data $p_T$

**Figure 6.33.** Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $p_T$

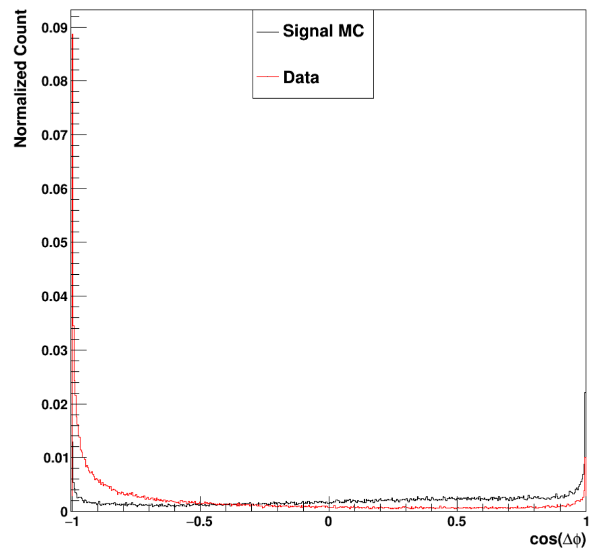
The scale of the accepted $p_T$ values is quite skewed by the very large bin content of the overflow bin (above 200 GeV/c). This points to another important kinematic difference – given the more central photon distribution in $\eta_{SC}$, the photon candidates tend to have higher $p_T$ values on average.

### 6.2.5 Creation of Diphoton BDT for $HH \to b\bar{b}\gamma\gamma$

The final step is to consider the recreation of the diphoton BDT for the DiHiggs process. It will be shown in figures 6.34-6.39 that both the kinematic variables and the input BDT score distributions are significantly different relative to the single Higgs.



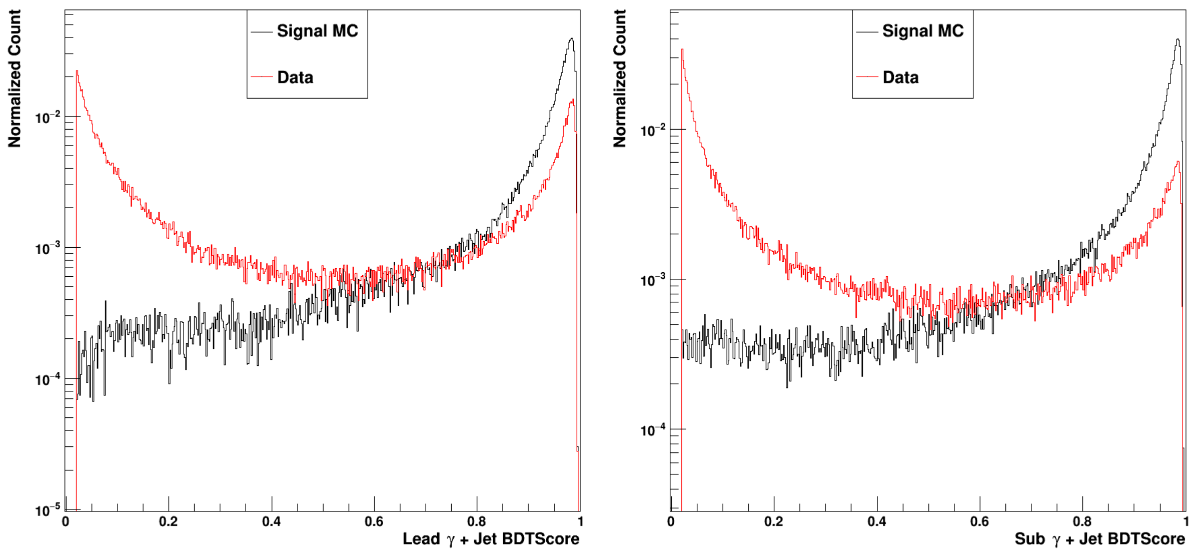

(a) $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$




(b) $\cos(\Delta\phi)$

**Figure 6.34.** The two per-event diphoton input variables,$\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ and $\cos(\Delta\phi)$. The signal MC is in black, whereas the data events are in red.

While the $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ disributions are relatively similar to their single Higgs counterparts, there are minor differences in the cos($\Delta\phi$). The single Higgs for both the data and signal MC was strongly peaked at -1, with a very similar shape for both samples. In the DiHiggs, however, there is far better separation between the signal and background; the data is still relatively well clustered at -1, but the signal MC has a sloped shape favoring higher values (especially increasing approaching 1). This, again, is due to the kinematics of the DiHiggs production. Cos($\Delta\phi$) = -1 corresponds to a $\Delta\phi = \pi$, whereas Cos($\Delta\phi$) = 1 corresponds to a $\Delta\phi = 0$. As expected, then, the DiHiggs signal is more likely to have a lower separation in $\phi$. So, the discrimination power of this variable should have increased relative to the single Higgs.

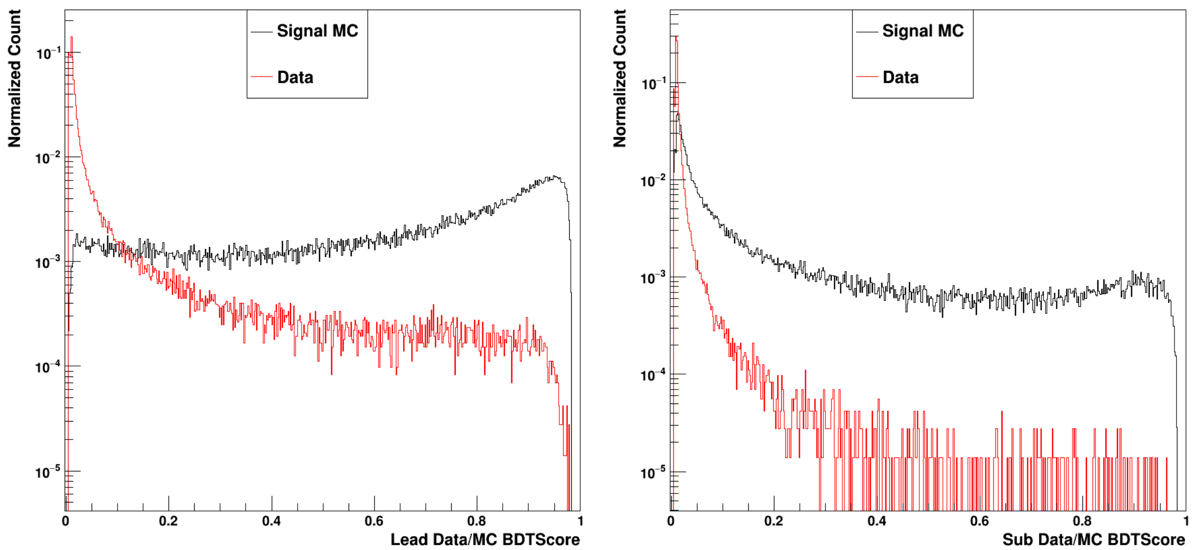



**(a)** Lead $\gamma$ + Jet BDT Score

**(b)** Sub $\gamma$ + Jet BDT Score

**Figure 6.35.** The lead and sublead $\gamma$ + Jet XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

Next, there are clear differences in the $\gamma$ + Jet BDT score distributions for the DiHiggs. The signal MC, surprisingly, has a relatively similar distribution in scores. The score peaks strongly at 1 for the signal in both the lead and sublead photon candidate. The data, however, has moved closer to 0. Especially in the sublead photon candidate, the peak near 0 is taller than the peak near 1. In the single Higgs, the two peaks were of a more similar size. Even though this $\gamma$ + Jet BDT was mot retrained, it appears to perform better on separating DiHiggs signal MC from data.

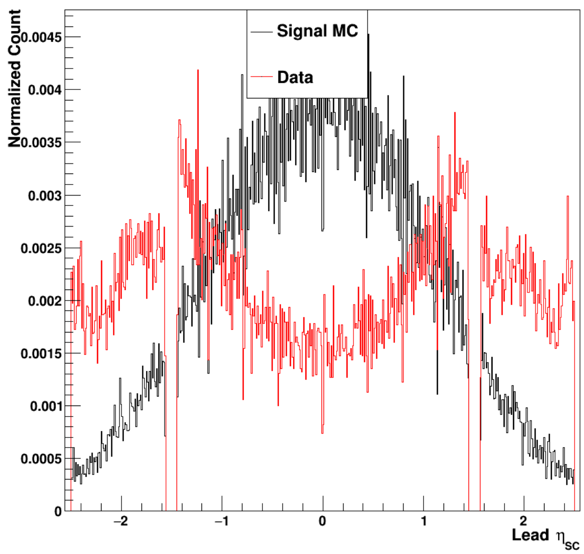



**(a)** Lead Data/MC BDT Score

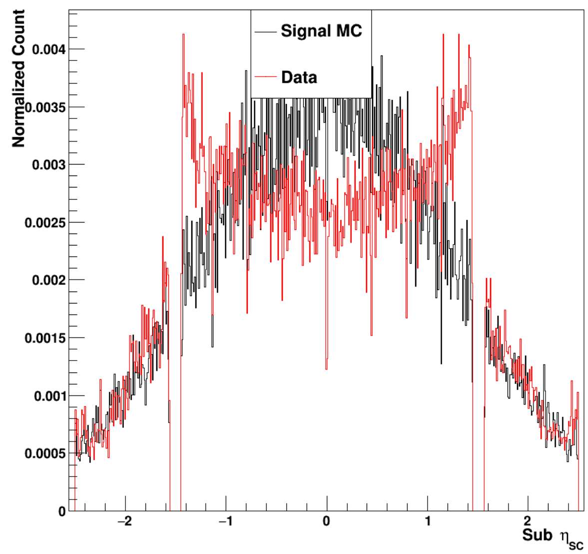
**(b)** Sub Data/MC BDT Score

**Figure 6.36.** The lead and sublead Data/MC XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

The single photon Data/MC BDT scores are quite well-separated for the lead photon, perhaps moreso than they were for the single Higgs sample. Clearly, as well, the sublead scores for the data are very well-placed near 0. Again, in conjunction with the $\gamma +$ Jet scores shown in figure 6.35, these data/MC BDT scores will have strong discrimination power.



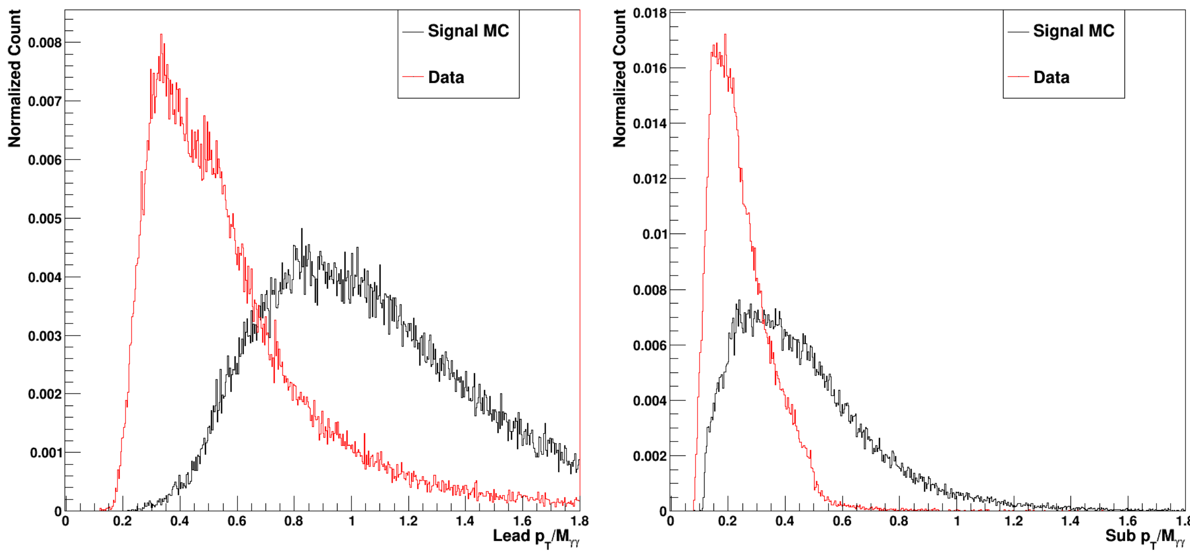

**(a)** Lead $\eta_{SC}$




**(b)** Sub $\eta_{SC}$

**Figure 6.37.** The lead and sublead $\eta_{SC}$. The signal MC is in black, whereas the data events are in red.

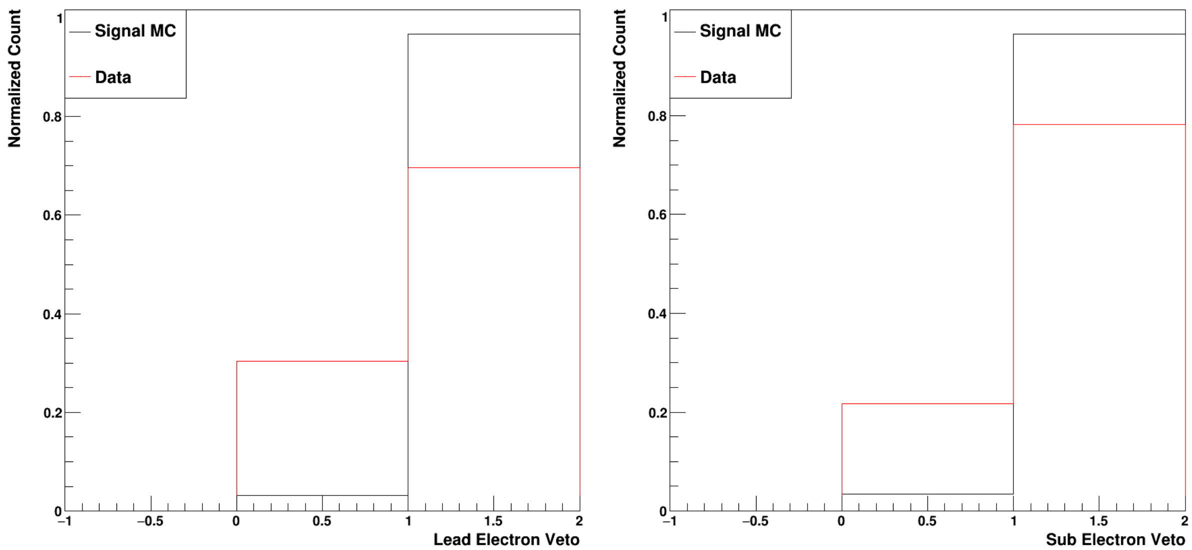
As mentioned for the selection of the single photon cuts, the $\eta_{SC}$ distributions for the diHiggs samples are more separated than those of the single Higgs. Especially in the lead photon candidate, there is very little data in the center of the barrel, while the signal MC sees its strongest peak in this region. The endcap shapes are rather different for the lead photon, but much more similar for the sublead photon. Again, due to the difference in kinematics relative to the single Higgs, these differences are to be expected.




**(a)** Lead $p_T/M_{\gamma\gamma}$ **(b)** Sub $p_T/M_{\gamma\gamma}$

**Figure 6.38.** The lead and sublead $p_T/M_{\gamma\gamma}$. The signal MC is in black, whereas the data events are in red.

While the separation between the signal and background for both lead and sublead $p_T/M_{\gamma\gamma}$ is still quite good (as seen in figure 6.38), the distributions for the signal MC are quite different than those for the single Higgs. Specifically, the values tend to be, on average, a bit higher for both the lead and sublead. Additionally, the spread of the values for the signal MC is quite a bit higher, which likely reduces the discrimination power of the $p_T/M_{\gamma\gamma}$ for the diHiggs samples.




**(a)** Lead Electron Veto **(b)** Sub Electron Veto

**Figure 6.39.** The lead and sublead electron veto The signal MC is in black, whereas the data events are in red.

Finally, the electron veto values look very similar to those of the single Higgs data and MC. Shown in figure 6.39, it is still the case that the signal MC has both photons passing the veto in >95% of cases. Again, this will prove a very useful discrimination variable.

For this final BDT, the same steps of hyperparameter optimization and training were followed. It was found that, for the diHiggs diphoton BDT, the same hyperparameters could be used as for the single Higgs version of the same BDT.